UNIVERSIDAD POLITÉCNICA DE MADRID

ESCUELA TÉCNICA SUPERIOR DE INGENIEROS DE TELECOMUNICACIÓN

MASTER OF SCIENCE IN NEUROTECHNOLOGY

MASTER´S THESIS

DESIGN, IMPLEMENTATION AND SYSTEMATIC EVALUATION OF PREPROCESSING PIPELINES FOR SILENT SPEECH DECODING IN ELECTROENCEPHALOGRAPHY

A STUDY ON DC COMPONENTS AND PREPROCESSING EFFECTS IN BCI2020

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Master of Science in Neurotechnology

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Title: Design, Implementation and Systematic Evaluation of Preprocessing Pipelines for Silent Speech Decoding in Electroencephalography

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A mis padres, por estar siempre al otro lado del teléfono
para escucharme y recordarme que nunca estoy solo

Agradecimientos

Quiero comenzar agradeciendo profundamente a mi familia, y en especial a mis padres, que han estado siempre a mi lado en cada paso del camino. Gracias por impulsarme y cuidarme incluso sin entender del todo lo que hacía, por estar siempre al otro lado del teléfono para escucharme, y por apoyarme con gestos sencillos pero inmensos, como traerme la comida o distraerme cuando más lo necesitaba. Gracias por recordarme, con su presencia constante, que nunca he estado solo. También quiero dar las gracias a mi hermano, que siempre ha estado dispuesto a tenderme la mano, a escucharme y aconsejarme con paciencia y cariño. Gracias por hacerme reír cuando más lo necesitaba, por inspirarme con tu forma de ver la vida y por demostrarme que el apoyo verdadero se construye día a día, con gestos sinceros y constantes.

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Jorge de la Rosa Padrón
Madrid, 2025

Resumen

El habla silenciosa se refiere al reconocimiento de frases y palabras que una persona intenta producir sin generar sonido audible. Esta línea de investigación ha atraído la atención considerablemente debido a su potencial para restaurar la comunicación en individuos con discapacidades motoras o del habla. En esta tesis nos centramos en la decodificación basada en electroencefalografía. Un desafío clave radica en la baja relación señal–ruido que tiene, lo que tradicionalmente ha motivado el uso de medios de preprocesado. Sin embargo, en los últimos años varios estudios han reportado resultados competitivos utilizando datos en bruto sin preprocesar, lo que genera preocupaciones de que los modelos de aprendizaje automático puedan explotar inadvertidamente artefactos no neuronales en lugar de correlatos genuinos del habla silenciosa. Utilizando el conjunto de datos *BCI Competition 2020 Track 3*, realizamos una investigación de cómo la actividad de muy baja frecuencia, y en particular la componente de corriente continua, influye en el rendimiento de la decodificación a través de múltiples arquitecturas de redes neuronales. Nuestros análisis demuestran que la componente DC es la principal fuente de información en este conjunto de datos. Cuando se conserva, incluso clasificadores sencillos alcanzan precisiones comparables a las reportadas por modelos de aprendizaje profundo. Cuando se elimina, el rendimiento cae drásticamente, lo que indica que los resultados previamente reportados pueden estar fuertemente influenciados por desplazamientos DC dependientes de la clase. Aunque este efecto es consistente y sistemático en todos los sujetos, la evidencia disponible no permite determinar si refleja una señal fisiológica genuina o un artefacto introducido durante la adquisición. De forma relevante, al evaluar un corpus independiente no observamos los mismos patrones de DC dependientes de la clase, lo que sugiere que el fenómeno podría ser específico del conjunto BCI Competition 2020 Track 3. Sin embargo, dado que esta comparación se limita a un único conjunto adicional, se necesitan más replicaciones antes de establecer conclusiones definitivas sobre su generalidad. Estos resultados ponen de manifiesto una dependencia consistente del componente DC en los modelos, lo que sugiere que desempeña un papel central en el rendimiento de la clasificación. No obstante, su naturaleza permanece sin resolver: podría reflejar un correlato fisiológico genuino del habla silenciosa o, alternativamente, surgir de propiedades sistemáticas del proceso de grabación. Independientemente de su origen, el fenómeno resulta científicamente significativo, ya que revela cómo los modelos pueden extraer estructura de la actividad de muy baja frecuencia y motiva investigaciones futuras para determinar si dicha información representa una señal neuronal significativa o un artefacto.

Palabras clave

Decodificación de habla silenciosa, electroencefalografía, preprocesado, componente DC, BCI Competition 2020 Track 3, aprendizaje profundo, transformers

Summary

Silent speech decoding refers to the recognition of words and phrases that a person intends to produce without generating audible sound. This line of research has attracted considerable attention due to its potential to restore communication in individuals with severe motor or speech impairments. In this thesis we focus on electroencephalography-based decoding, where electrical potentials recorded at the scalp are used to infer covert speech processes. A key methodological challenge lies in the low signal-to-noise ratio of electroencephalography, which has traditionally motivated extensive preprocessing pipelines including band-pass filtering, baseline correction, and artifact removal. However, in recent years several studies have reported competitive results on raw, unprocessed data, raising concerns that machine learning models may inadvertently exploit non-neural artifacts rather than genuine correlates of silent speech. Using the BCI Competition 2020 Track 3 dataset, we carried out a systematic investigation of how ultra-low-frequency activity, and in particular the direct current component, influences decoding performance across multiple neural network architectures. Our analyses demonstrate that the DC offset is the primary source of discriminative information in this dataset. When this component is retained, even simple classifiers achieve accuracies comparable to those reported by state-of-the-art deep learning models. When it is removed, performance drops dramatically, indicating that previously reported findings may in fact be strongly influenced by class-dependent DC offsets rather than by unequivocal neural correlates of silent speech. While this effect is consistent and systematic across subjects, the present evidence does not allow us to determine whether it reflects a genuine physiological signal or an artifact introduced during data acquisition. Importantly, when evaluating an independent corpus we did not observe the same class-dependent DC patterns, which suggests that the phenomenon may be specific to the BCI Competition 2020 Track 3 dataset. However, given that this comparison is limited to a single additional dataset, further replication is needed before drawing definitive conclusions about its generality. These results highlight a consistent reliance on the DC component across models, suggesting that it plays a central role in driving classification performance. At present, however, its precise nature remains unresolved: it could reflect a genuine physiological correlate of silent speech, or alternatively arise from systematic properties of the recording process. Regardless of its origin, the phenomenon is scientifically significant, as it reveals how models can extract discriminative structure from ultra-low-frequency activity, and it motivates further research to determine whether such information represents a meaningful neural signal or an artifact.

Keywords

Silent speech decoding, electroencephalography, preprocessing, ultra-low frequency artifacts, DC offset, BCI Competition 2020 Track 3, deep learning, transformer models

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Glossary

ALS	Amyotrophic Lateral Sclerosis
BCI	Brain–Computer Interface
CAE	Conditional Autoencoder
CAR	Common Average Reference
CFC	Cross-Frequency Coupling
CNN	Convolutional Neural Network
CSP	Common Spatial Patterns
DDPM	Denoising Diffusion Probabilistic Model
ECoG	Electrocorticography
ECG	Electrocardiography
EEG	Electroencephalography
EMG	Electromyography
EOG	Electrooculography
ERD	Event-Related Desynchronization
ERP	Event-Related Potential
ERS	Event-Related Synchronization
FBCSP	Filter Bank Common Spatial Patterns
FIR	Finite Impulse Response
HPC	High-Performance Computing
ICA	Independent Component Analysis
IFG	Inferior Frontal Gyrus
M1	Primary Motor Cortex
MEG	Magnetoencephalography
MLP	Multilayer Perceptron
SDG	Sustainable Development Goals
SE	Squeeze–Excitation
SMA	Supplementary Motor Area
SNR	Signal-to-Noise Ratio
SSD	Silent Speech Decoding
STG	Superior Temporal Gyrus
STS	Superior Temporal Sulcus
SVM	Support Vector Machines
ViT	Vision Transformer

Introduction and objectives

1.1 Context

Human speech is one of the most complex and coordinated acts performed by our nervous system. What appears to us as an effortless act like uttering a single word, is in fact the outcome of a long sequence of neural processes that begin well before any muscle moves. It all starts with an intention: the mental decision to communicate a message. This intention triggers neural networks responsible for selecting the appropriate word, assembling its phonemes in the correct order, and planning the movements of the lips, tongue, and larynx required to produce it.

From a neurophysiological perspective, speech production begins with the planning of the message in the prefrontal cortex and temporal areas, followed by phonological encoding in regions such as Wernicke's and Broca's areas, and finally by the motor programming of speech in premotor and motor cortices, including the supplementary speech area. Within fractions of a second, this plan travels from the cortical language and motor areas to the cranial nerves that innervate the speech muscles. These muscles then contract in precise coordination with respiration and vocal fold vibration to transform the intended word into sound. Yet, this process does not necessarily proceed all the way to sound production. Communication can remain "halfway": neural activity may be present without muscular movement, or motor commands may be issued without airflow through the larynx to generate sound. It is in these intermediate states that electroencephalography (EEG) and electromyography (EMG) become particularly relevant: EEG captures cortical activity, whereas EMG registers muscle activation.

Silent speech decoding (SSD) is therefore the recognition of a person's intended speech without producing audible sound. It is a branch of brain–computer interfaces (BCIs) aimed at identifying words, phrases, or phonemes that a person seeks to produce in the absence of audible speech. For many individuals, speech is not something to be taken for granted. Neurodegenerative diseases such as amyotrophic lateral sclerosis (ALS), stroke, spinal cord injuries, or conditions such as locked-in syndrome can severely impair or eliminate the ability to communicate verbally. For these individuals, SSD offers a pathway to regain functional communication.

Two main classification paradigms exist in SSD, depending on the vocabulary scope:

(i) Closed-vocabulary: the system is trained to discriminate among a fixed and limited set of words or phrases.

(ii) Open-vocabulary: the system is expected to recognize any word, including those unseen during training, enabling more precise and flexible expression.

In this thesis, we will focus on decoding based solely on EEG signals, corresponding to the planning and motor execution stages of speech and without muscular articulation, on the closed-vocabulary paradigm, which currently offers higher accuracy and practical utility for patients.

1.2 Impact of Preprocessing on EEG Decoding

EEG measures the brain's electrical activity at the scalp. By the time cortical potentials reach the electrodes, they are typically attenuated, spatially mixed, and contaminated by multiple non-neural sources. Common physiological artifacts include eye blinks, saccades, and facial muscle activity; additional contamination often arises from environmental factors such as electrode motion, impedance changes, or mains interference. The combined effect is a low signal-to-noise ratio (SNR), which has traditionally motivated the use of preprocessing pipelines to render EEG signals more amenable to analysis.

Preprocessing refers to the set of operations applied prior to feature extraction or model training. In practice, this often includes band-pass filtering to remove slow drifts and high-frequency noise, notch filtering to suppress line noise, artifact handling, re-referencing and amplitude normalization to improve channel comparability, and baseline correction or detrending to suppress slow, task-unrelated trends. The conventional assumption is that these steps enhance signal quality and, consequently, downstream decoding performance.

However, this assumption does not always hold. Some studies have reported that certain preprocessing choices can reduce classification performance, and that the effect is model- and task-dependent. For example, Kessler et al. [1] observed that several artifact-correction procedures degraded accuracy for architectures such as EEGNet and temporal regressors, whereas increasing the high-pass cutoff improved results; by contrast, operations like detrending or baseline correction showed mixed effects depending on the model and paradigm. Del Pup et al. [2] indicated that aggressive artifact-correction stages can consistently depress decoder performance, while raising the low-pass cutoff can yield improvements; furthermore, Defossez et al. [3] showed that more elaborate pipelines have not always outperformed simpler baselines (e.g., baseline combined with robust scaling).

Recent work with the BCI2020 Track 3 dataset [4] illustrates an important methodological issue: several studies have reported results obtained on raw EEG recordings without any form of preprocessing. This trend may reflect the growing confidence in deep neural networks as end-to-end learners, inspired by domains such as natural language processing, where early handcrafted pipelines have been replaced by directly feeding raw text to large models. However, EEG is fundamentally different: its low signal-to-noise ratio and the presence of structured artifacts have traditionally motivated the removal of very slow components, under the assumption that they mainly reflect noise rather than task-related neural activity.

Yet, by omitting such preprocessing, models may end up relying on signal components whose physiological relevance remains uncertain. In other words, the absence of filtering does not guarantee access to more genuine neural information; instead, it opens the possibility that decoding performance is driven by structures that could equally represent recording artifacts or slow neural processes, the exact nature of which is still unresolved.

This thesis is motivated by that observation. In BCI2020 Track 3, we show that not only components below 0.05 Hz (hereafter referred to as ultra-low frequencies, ULF), typically removed by standard band-pass filters, but even the DC offset itself are sufficient to drive classification performance. In fact, a simple support vector machine (SVM) trained exclusively on the DC component reaches accuracies comparable to those reported in the literature (around 51%). These findings indicate that a large portion of the decoding performance in this dataset can be explained by systematic DC offsets, although the evidence available does not yet allow us to establish whether they represent a physiological signal or a recording-related effect. While prior studies often adopted fixed preprocessing pipelines without examining these issues, here we systematically contrast raw and preprocessed variants across multiple architectures. Our goal is to clarify how strongly models depend on it and to provide a transparent basis for future work. While the present evidence does not allow us to determine its precise nature, we discuss several plausible hypotheses. Clarifying which of these explanations holds true remains an open and important direction for subsequent research.

1.3 Objectives

The primary objective of this work is to examine the methodological implications of omitting preprocessing in silent speech EEG decoding. In particular, we aim to demonstrate that in the widely used BCI2020 Track 3 dataset, high decoding accuracies reported on raw data may not only be shaped by ultra-low-frequency components below 0.05 Hz, but can even be largely accounted for by the DC offset itself. Remarkably, simple classifiers are able to exploit this offset to reach performance levels comparable to those of state-of-the-art models, suggesting that a substantial part of decoding accuracy in this dataset depends on slow components whose precise nature remains unresolved. By systematically contrasting raw and preprocessed variants across eight different architectures, we seek to clarify to what extent the absence of standard filtering shifts model performance toward such low-frequency structures, and what this implies for the interpretation of reported results.

A secondary objective is to conduct a systematic comparison of preprocessing configurations, including standard band-pass filtering, baseline correction, and partial ablations, in order to quantify their relative impact on model performance. Beyond reporting accuracy numbers, our goal is to provide practical methodological guidance: identifying which preprocessing steps most strongly affect decoding outcomes, and characterizing why and how models learn to rely on the DC component in particular. This analysis is intended to support a clearer understanding of the role of DC components in silent speech decoding, and to guide future studies in evaluating whether such information reflects meaningful neural signals or recording-related influences.

Hypotheses

Based on prior work and our preliminary findings, we formulate the following hypotheses:

H1: Classification performance obtained from raw BCI2020 Track 3 data is strongly influenced by ultra-low-frequency activity, and in particular by the DC component, leading to unexpectedly high accuracies.

H2: Standard preprocessing pipelines (e.g., band-pass filtering, baseline correction) reduce the contribution of these slow components, resulting in lower but potentially more physiologically grounded decoding performance.

H3: The impact of preprocessing on performance is expected to depend on both the dataset characteristics and the specific architecture, leading to variations in how strongly models rely on low-frequency information.

H4: The DC component can be systematically characterized and shown to play a central role in model decisions, whether it is physiological or recording-related.

1.4 Structure of the Document

The thesis is organised into seven chapters, each addressing a specific aspect of the work. Chapter 2 reviews the theoretical background and situates the present study within the field of brain–computer interfaces. It introduces the neurophysiological basis of imagined and silent speech, discusses common preprocessing pipelines in electroencephalography, and summarises the main deep learning architectures that have been applied to EEG decoding, with particular emphasis on recent diffusion-based approaches.

Chapter 3 describes the materials and methodology adopted in this work. It details the characteristics of the BCI Competition 2020 Track 3 dataset, the preprocessing configurations considered, and the model architectures evaluated, including proposed variants of Diff-E. It also explains the experimental protocols designed to isolate the contribution of ultra-low-frequency components, together with the computational environment employed.

Chapter 4 presents the results of the empirical analyses together with their interpretation. It covers comparisons between raw and preprocessed signals, ablation studies of different preprocessing steps, and the isolation of ultra-low-frequency activity with a particular focus on the

DC component. The chapter also examines how different model families exploit these slow signal structures. Throughout, the findings are interpreted in relation to existing literature, with attention to their implications for the field of silent speech decoding, possible explanations for the observed phenomena, and the methodological lessons that can be drawn.

Chapter 5 synthesises the main contributions of the thesis, acknowledges its limitations, and suggests future research directions.

Finally, Chapter 6 addresses the ethical, societal, economic, and environmental considerations relevant to this work, and Chapter 7 provides a financial budget. The document concludes with a bibliography of the cited works and annexes containing complementary material.

State of the Art

2.1 Neurophysiological basis of imagined speech EEG

As outlined in Chapter 1, speech production engages a distributed cortico–cortical network that plans, encodes and prepares utterances on sub-second timescales. Two complementary theoretical frameworks organise this network. First, dual-stream models posit a ventral stream for mapping acoustic/phonological representations onto meaning and a dorsal stream for sensorimotor mappings between auditory/phonological codes and articulatory motor programs [5], implicating superior temporal cortex (STG/STS), inferior frontal gyrus (IFG/Broca's area), inferior parietal areas, premotor cortex, supplementary motor area (SMA) and primary motor cortex (M1). These circuits remain relevant even without overt articulation, because internal phonological and articulatory representations are recruited during covert speech [6, 7].

Second, motor–sensory internal-model accounts propose that speech preparation generates forward predictions of the sensory consequences of planned articulatory commands via corollary discharge/efference copy [8, 9]; these predictions modulate auditory and somatosensory cortices during both overt and imagined speech. Neurophysiological evidence from MEG, EEG or ECoG shows temporally precise signatures of such predictive signals during inner speech and motor imagery, consistent with hierarchical monitoring in auditory cortex [10].

These frameworks make concrete predictions for scalp EEG. In sensorimotor regions, μ (8–13 Hz) and β (13–30 Hz) rhythms typically exhibit event-related desynchronization (ERD) during preparation and imagery of movement, including articulatory gestures. Speech EEG work shows that μ/β dynamics, often localized over pericentral electrodes, index sensorimotor engagement of the vocal tract even without overt movement, and Independent Component Analysis (ICA) can help isolate these sources [11, 12]. Beyond low frequencies, invasive recordings consistently report high-γ (70–150 Hz) increases over IFG, premotor/M1 and peri-Sylvian cortex during speech production, reflecting local population firing; while high-γ is attenuated on the scalp, preserving upper bands can still be informative [13, 14, 15].

For imagined speech specifically, recent intracranial work indicates that low-frequency power and cross-frequency interactions (e.g., θ/γ, β/γ) carry key information for phonetic–vocalic structure, whereas overt speech is dominated by high-γ [16]. This supports EEG pipelines that retain low-frequency dynamics while avoiding overly aggressive low-pass filtering that could disrupt relevant cross-frequency structure [17, 18, 19].

In summary, the neurophysiology motivates three design guidelines we leverage in this work: to retain canonical frequency bands (δ–β) associated with sequencing, phonological maintenance, and motor preparation; to preserve high-γ activity, which dominates overt speech production and provides fast local dynamics as well as cross-frequency interactions, avoiding excessive low-pass filtering that could suppress these signals; and to apply referencing and artefact handling strategies that minimize global or non-neural variance while preserving the subtle, distributed effects expected in imagined speech. These principles directly inform the preprocessing comparisons and the architectural choices evaluated below.

2.2 EEG preprocessing pipelines

Preprocessing is a foundational stage in EEG analysis, aiming to improve the relation between raw neural recordings and the features ultimately used by machine learning models. While Chapter 1 introduced the implications surrounding preprocessing in silent speech decoding, here we broaden the scope to its technical and historical background.

From the earliest EEG-based brain–computer interfaces, scalp recordings have been recognised as noisy and artefact-prone, requiring substantial preparation before analysis [20]. The design of a preprocessing pipeline typically depends on the intended application, the acquisition hardware, and the statistical or computational model employed. Standard operations include temporal filtering, artefact attenuation or rejection, spatial transformations, and amplitude normalisation. However, the sequence, parameterisation, and even the necessity of some of these steps remain the subject of active debate [21]. The following sections review the main categories of preprocessing operations:

Notch filtering

A notch filter is a narrow-band stop filter designed to attenuate a specific frequency (and optionally its harmonics) while leaving neighbouring frequencies relatively unaffected. In EEG preprocessing, the most common application is the suppression of electrical mains interference. In Europe this typically occurs at 50 Hz, whereas in North America (or South Korea) it is centred at 60 Hz. This interference arises from electromagnetic coupling between the EEG acquisition system and the surrounding electrical infrastructure, including power lines, fluorescent lighting, and electronic equipment. Its presence can mask or distort neural oscillations in overlapping frequency bands.

Band-pass filtering

A band-pass filter attenuates signal components outside a specified frequency range, preserving only those frequencies of interest for the analysis. In EEG preprocessing, this serves two complementary purposes: the high-pass component suppresses slow drifts caused by factors such as electrode–skin impedance changes, perspiration, or movement-related baseline shifts; the low-pass component removes high-frequency noise, including muscle activity, main harmonics, and sensor/electronic noise. The chosen passband is typically guided by the spectral content of cortical oscillations relevant to speech production. Classical EEG frequency bands provide a useful reference[1] :

- δ (Delta, 0.5–4 Hz): associated with slow cortical potentials, often linked to global attention and timing processes.
- θ (Theta, 4–8 Hz): implicated in working memory and sequential processing, relevant for phonological encoding.
- α (Alpha, 8–12 Hz): associated with cortical inhibition and sensorimotor idling.
- β (Beta, 13–30 Hz): strongly modulated during speech motor planning and execution, particularly in sensorimotor cortices.
- γ (Gamma, >35 Hz): reflects high-frequency local synchrony, potentially capturing fine-grained motor or linguistic processes.

While these bands provide a conceptual scaffold, exact cut-off frequencies vary between studies, depending on hypotheses, recording hardware, and noise characteristics.

Re-referencing

EEG measures the potential difference between each recording electrode and a designated reference electrode. Because the reference is itself an active measurement site, its choice directly influences the observed amplitudes, spatial distributions, and even the apparent timing of EEG features. Re-referencing is the process of mathematically redefining this reference after acquisition to improve signal interpretability and reduce reference-specific biases. One of the most widely used schemes is the Common Average Reference (CAR), in which the mean potential across all (or a subset of) electrodes is subtracted from each channel. The rationale is that potentials common to all electrodes (such as global noise or slow drifts) will be attenuated, enhancing spatial specificity.

Independent Component Analysis

ICA is a blind source separation technique that decomposes multichannel EEG data into a set of statistically independent components. The rationale is that the observed scalp potentials are mixtures of underlying neural and non-neural sources; by unmixing them, it becomes possible to identify and selectively remove artefactual sources while preserving the remaining neural signal. In EEG preprocessing, ICA is widely used to detect and suppress components corresponding to ocular activity (blinks, saccades; EOG), cardiac signals (ECG), and muscle activity (EMG).

[1] EEG frequency bands are not defined by a universal standard. Reported cut-offs may vary slightly across studies (e.g., some extend β up to 35 Hz or start γ above 30 Hz). The chosen ranges here follow a common convention for clarity.

Baseline correction

Baseline correction is a common step in event-related EEG analysis, particularly for epoched data. The method involves subtracting the mean voltage of a predefined pre-stimulus interval from the entire epoch, re-centering the signal so that zero microvolts represents the baseline level. The traditional motivation is to eliminate DC offsets and slow drifts, thereby stabilising the voltage baseline prior to temporal or ERP analyses. A classic choice for the baseline window is –200 to 0 ms relative to stimulus onset, though exact intervals vary with paradigm and expected neural latencies.

As already noted in the Introduction (Section 1.2), several works report results on the BCI Competition 2020 Track 3 dataset obtained directly from raw EEG recordings, without the application of standard preprocessing pipelines such as bandpass filtering or baseline correction. For instance, Lee et al. [22] trained convolutional models on unfiltered signals using their proposed length-wise training strategy, with only the receptive fields of the first convolutional layer incidentally attenuating ocular activity. Other studies also adopt no preprocessing: Lee et al. [23] explored transfer learning from spoken to imagined speech EEG, directly applying models across paradigms; Ko et al. [24] compared machine learning and deep learning approaches using raw segmented windows; and Alharbi and Alotaibi [25] transformed raw EEG into topographic maps that were then processed by hybrid 3D-CNN–RNN architectures. In addition, Jiang et al. [26] reports results on a private dataset, although its public implementation relies on BCI2020 Track 3 and also ingest the data without any preprocessing [27]. Taken together, these examples highlight that omitting preprocessing is not a rare exception but rather a repeated methodological choice in recent studies on BCI2020 Track 3, underscoring the importance of carefully examining how such decisions influence the interpretation of reported results.

2.3 Deep learning architectures for EEG

The purpose of this section is not to benchmark architectures in search of the highest decoding accuracy. Rather, the aim is to examine how different model families behave under varied preprocessing conditions, and to what extent their reported performance may be driven by artifacts instead of genuine neural patterns. To this end, we include a set of widely used convolutional baselines, together with EEG-specific compact models and one recent diffusion-based approach. The rationale is that models with different inductive biases and representational capacities may either amplify or attenuate the influence of ULF artifacts when trained on raw BCI2020 Track 3 signals. By comparing their performance across systematically varied pipelines, we seek to highlight that the key issue is not architectural novelty per se, but the interaction between preprocessing (or its absence) and model sensitivity. The subsections below briefly outline each architecture and its relevance to imagined speech decoding.

2.3.1 Baseline architectures

The baseline models considered in this study are not tuned to achieve state-of-the-art accuracy, but rather to provide representative coverage of widely used paradigms in EEG decoding. Their prevalence in the literature makes them suitable for assessing whether preprocessing effects generalise across models of varying complexity.

CNN A convolutional neural network (CNN) for EEG decoding applies a set of learnable temporal filters over each electrode signal, transforming the input into feature maps through convolution and nonlinear activation. These first-layer temporal filters behave as finite impulse response (FIR) filters, often learning to emphasise frequency components that are informative for the task. In motor-related and some imagined-speech paradigms, it has been empirically observed that these learned filters frequently align with the μ[2] and β rhythms. Deeper convolutional layers combine temporal patterns and integrate spatial information across channels, while pooling operations reduce temporal resolution and improve robustness to small temporal shifts. Finally, a dense layer maps the flattened feature maps to class probabilities via a softmax function.

EEGNet EEGNet is a compact convolutional architecture widely adopted in BCI research due to its balance between interpretability, computational efficiency, and accuracy [29]. It was specifically designed to emulate the behaviour of classical filterbank spatial filtering methods, such as Common Spatial Patterns (CSP) or Filter Bank CSP (FBCSP), but with the advantage of learning both the spectral filters and the spatial projections directly from data in an end-to-end manner. Its design allows the model to find frequency bands and electrode patterns that maximise class separability, capturing physiologically meaningful rhythms without the need for handcrafted feature extraction.

ShallowNet ShallowConvNet [30] extracts oscillatory power features in a manner analogous to band-power computation. A long temporal convolution isolates activity in specific frequency bands, followed by a spatial filter combining signals across electrodes. A logarithmic transform stabilises variance and approximates Gaussianity, which benefits the subsequent linear classification stage. This physiologically motivated architecture facilitates direct mapping between learned filters and canonical EEG rhythms.

[2] Although it shares the same frequency range as the visual α rhythm, the μ rhythm originates in the sensorimotor cortex and attenuates during actual or imagined movement, whereas the visual α rhythm arises from the occipital cortex and attenuates upon eye opening [28].

DeepConv DeepConvNet [30] employs a hierarchy of convolution–activation–pooling blocks to capture increasingly abstract EEG features. The first block applies temporal filtering followed by spatial projection across electrodes, similar to ShallowConvNet. Subsequent convolutional layers extract progressively higher-order patterns, combining spectral, spatial, and temporal representations. Pooling reduces temporal resolution at each stage, while dropout regularises learning. Unlike ShallowConvNet, which focuses on oscillatory power extraction within predefined frequency ranges, DeepConvNet leverages multiple layers to learn richer, more abstract features, often at the expense of reduced interpretability for higher accuracy in complex EEG decoding tasks.

2.3.2 Diffusion-based Learning for Imagined Speech EEG Decoding

Decoding imagined speech from EEG signals is demanding due to low signal quality, high variability between and within subjects, and the limited availability of large, labeled datasets. These factors make it difficult for standard discriminative models to generalize well, as they often overfit noise and miss the complex patterns that unfold across time and electrodes. Diff-E was chosen for this work because it combines a generative denoising process with a classification objective, encouraging the model to learn latent features that are both robust to noise and relevant for the task.

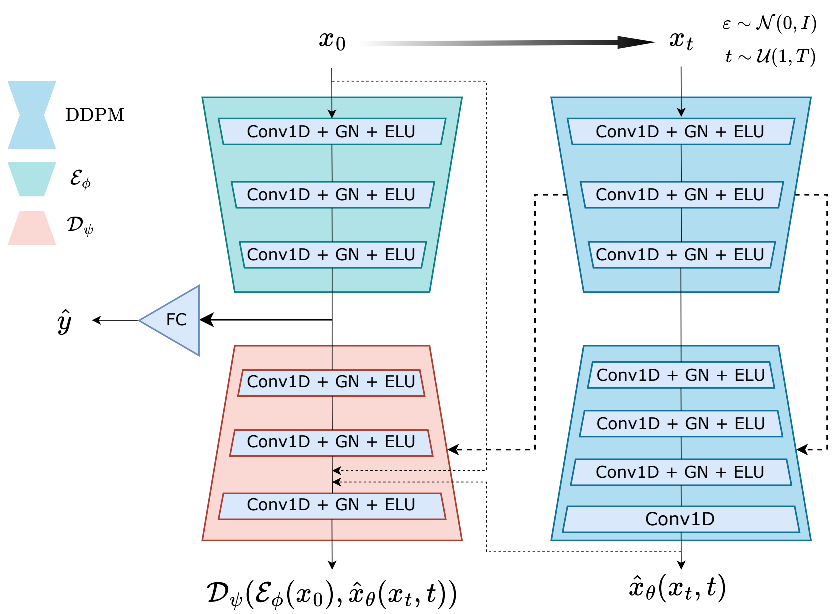

Figure 2.1: Architecture of the base Diff-E.

Diff-E builds upon the framework of denoising diffusion probabilistic models (DDPMs) [31], a family of latent-variable generative models that learn to approximate complex data distributions by reversing a fixed Markovian noising process. Formally, given a data distribution $q(x_0)$, the forward or diffusion process is defined as a sequence of Gaussian transitions:

$$q(x_t \mid x_{t-1}) = \mathcal{N}\Big(\sqrt{1-\beta_t}\, x_{t-1},\, \beta_t \mathbf{I}\Big), \quad t = 1, \ldots, T, \tag{2.1}$$

where $\{\beta_t\}_{t=1}^{T}$ is a predefined variance schedule with small positive values, and T is the total number of steps. This process progressively adds Gaussian noise to the initial clean sample x_0, such that x_T approaches an isotropic Gaussian distribution $\mathcal{N}(0, \mathbf{I})$.

By marginalizing over the intermediate variables $\{x_1, \ldots, x_{t-1}\}$, one can obtain a closed-form expression for sampling x_t directly from x_0:

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t)\mathbf{I}\big), \tag{2.2}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$ is the cumulative product of the α_s values up to step t. This property allows the generation of noisy samples at any arbitrary timestep t without iteratively applying the forward process.

The reverse process aims to invert this corruption and is modeled as:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(\mu_\theta(x_t, t),\, \Sigma_\theta(x_t, t)), \tag{2.3}$$

where θ are the learnable parameters of a neural network that predicts either the noise ϵ added at each step or the clean signal x_0. In the original DDPM formulation [31], the model $\epsilon_\theta(x_t, t)$ is trained to predict the Gaussian noise injected at step t, leading to the simplified loss function:

$$\mathcal{L}_{\text{DDPM}}^{\text{original}}(\theta) = \mathbb{E}_{t, x_0, \epsilon}\left[\|\epsilon - \epsilon_\theta(x_t, t)\|_2^2\right], \tag{2.4}$$

with x_t generated as:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \tag{2.5}$$

In contrast, Diff-E adopts a signal prediction parameterization: instead of predicting the noise ϵ, the network $\hat{x}_\theta(x_t, t)$ is optimized to directly estimate the clean sample x_0. This changes the training objective to:

$$\mathcal{L}_{\text{DDPM}}^{\text{Diff-E}}(\theta) = \mathbb{E}_{t, x_0}\left[\|x_0 - \hat{x}_\theta(x_t, t)\|_1\right], \tag{2.6}$$

where the L_1 loss encourages sharper reconstructions and is less sensitive to outliers compared to MSE. This parameterization also aligns better with the subsequent conditional autoencoder stage in Diff-E, which explicitly receives $\hat{x}_\theta(x_t, t)$ as input for further refinement. Thus, while the original DDPM learns to reverse the forward process in terms of noise estimation, Diff-E focuses on reconstructing the clean signal directly at each timestep. The DDPM backbone adopts a time-conditioned 1D UNet architecture, where temporal embeddings modulate the intermediate layers to encode the degree of corruption.

A key innovation of Diff-E is the introduction of a conditional autoencoder (CAE) that leverages the residual error patterns produced by the DDPM. During the forward diffusion process, certain fine-grained features of x_0 are inevitably lost; the CAE is trained to reconstruct these missing components by receiving as input both the original signal and the DDPM's denoised output, along with skip connections from the DDPM's internal feature maps. The CAE consists of an encoder E_ϕ and a decoder D_ψ, with the objective

$$\mathcal{L}_{\mathrm{CAE}}(\psi, \phi) = \left\| \mathcal{L}_{\mathrm{DDPM}}(\theta) - D_\psi\big(E_\phi(x_0), \hat{x}_\theta(x_t, t)\big) \right\|_1. \tag{2.7}$$

Skip connections from the DDPM to D_ψ implicitly condition the reconstruction on the diffusion timestep, enabling the CAE to focus on timestep-specific errors. Additionally, x_0 and $\hat{x}_\theta(x_t, t)$ are concatenated into the last layer of D_ψ for refined correction.

The discriminative component is seamlessly integrated into this pipeline. The encoder E_ϕ compresses the EEG sequence into alatent vector z by collapsing the temporal axis via adaptive average pooling. This vector is fed to a linear classifier C_ρ to predict the imagined speech class. The complete Diff-E loss function combines reconstruction and classification terms:

$$\mathcal{L}_{\mathrm{DiffE}}(\theta, \phi, \psi, \rho) = \left\| \mathcal{L}_{\mathrm{DDPM}}(\theta) - D_\psi\big(E_\phi(x_0), \hat{x}_\theta(x_t, t)\big) \right\|_1 + \alpha \left\| y - C_\rho(z) \right\|_2^2, \tag{2.8}$$

with $\alpha = 0.1$ balancing the generative and discriminative objectives. During inference, only E_ϕ and C_ρ are used, making the model efficient once trained.

A central motivation for combining a generative diffusion model with a secondary network that learns from its reconstruction errors is grounded in recent findings that diffusion models implicitly encode class-relevant structure in their denoising predictions, even without explicit discriminative training [32]. In this framework, the DDPM serves as a general signal model: by learning to reverse the forward noising process, it captures the dominant, class-agnostic structure of the EEG data distribution. The conditional autoencoder, in turn, focuses on predicting the residual difference between the DDPM's output and the original signal. This residual contains information not well explained by the generative model, mainly the fine-grained, class-specific variations that remain after the "average" or common structure of the signal has been reconstructed. From this perspective, the DDPM can be seen as modeling the shared components and stochastic variability of the EEG, while the encoder isolates the discriminative cues embedded in the deviation from that shared representation. This two-stage process encourages the latent representation to emphasize subtle, task-relevant features that might be lost in a purely generative or purely discriminative approach.

2.4 Summary and position of this work

The review above highlights an open methodological question that motivates this study. While EEG preprocessing is commonly applied to suppress slow drifts and artifacts, its omission is relatively frequent in recent imagined speech work with BCI2020 Track 3. Several studies report competitive results obtained directly on raw recordings, yet without clarifying whether the observed performance reflects genuine neural activity or systematic low-frequency components of uncertain origin. This ambiguity motivates a closer examination of what information models are exploiting when preprocessing is absent.

Accordingly, the present work shifts the focus from architecture benchmarking to a systematic investigation of how preprocessing choices shape model behaviour on BCI2020 Track 3. We employ established baselines (CNN, EEGNet, ShallowConvNet) together with more recent proposals such as diffusion-based approaches (Diff-E), not with the aim of identifying the best-performing architecture, but to assess their robustness under different preprocessing configurations. Particular attention is given to ultra-low-frequency and DC components, which we show can support high decoding accuracies even for simple classifiers such as SVMs. This suggests that unfiltered pipelines may capture slow structures that carry discriminative information, though whether these reflect neural processes or recording-related factors remains unresolved.

By grounding the analysis in a publicly available dataset and replicating common practices reported in the literature, this work positions itself as a methodological contribution. Its central message is not to privilege one architecture over another, but to highlight that performance in silent speech decoding can strongly depend on very slow signal components whose nature is still uncertain. In this sense, the study aims to clarify how models learn from these components, to motivate further research into their physiological versus non-physiological origins, and to encourage transparent reporting of preprocessing choices in EEG-based silent speech decoding.

Materials and Methods

3.1 Dataset Description

Public EEG datasets suitable for BCI research are scarce. In particular, datasets for imagined or covert speech decoding are extremely limited due to the sensitive nature of medical and neural data, which often remain restricted within the collecting institution. Ethical concerns, subject privacy, and the cost and complexity of EEG acquisition contribute to the low availability of high-quality, publicly released datasets. This scarcity forces many research groups to work with small, locally collected datasets, limiting reproducibility and the fair benchmarking of new algorithms.

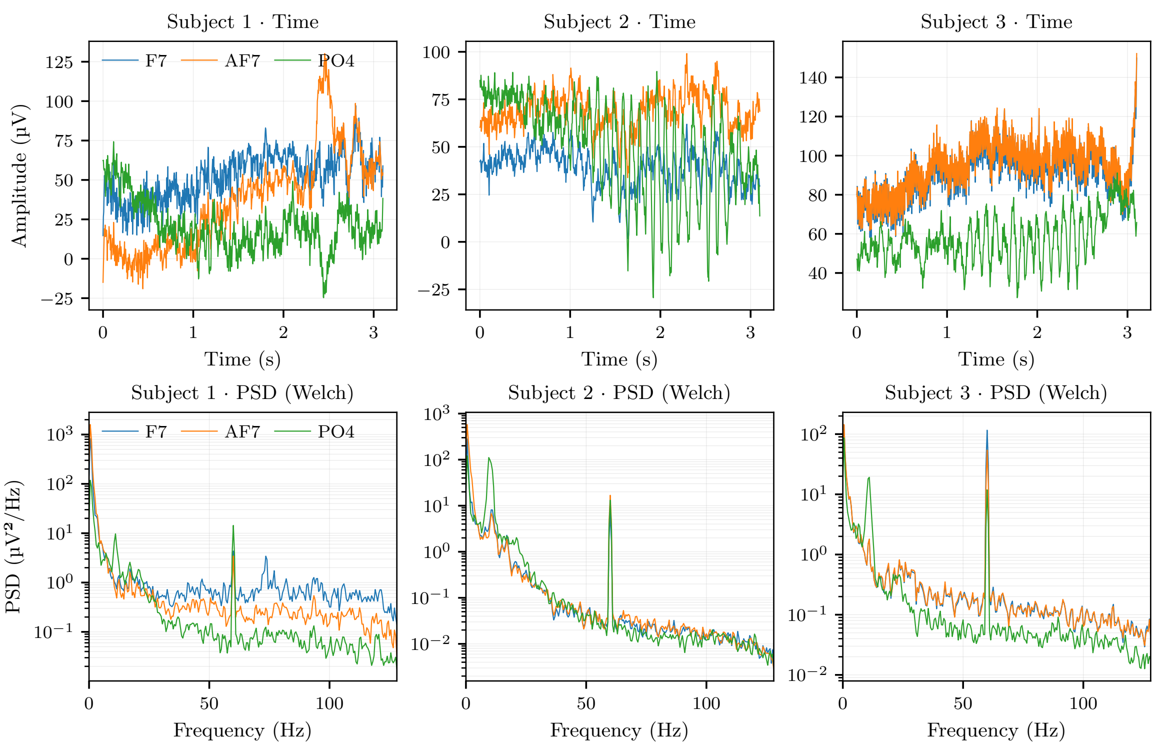

Figure 3.1: EEG examples from three subjects showing time signals (top) and PSDs (bottom) for channels F7, AF7, and PO4. No clear preprocessing is observed, as low-frequency components below 0.5 Hz and power-line artifacts around 60 Hz are still present.

In this work, we used the BCI Competition 2020, Track 3 dataset [4], released by the Department of Brain and Cognitive Engineering at Korea University (Seoul, South Korea) as part of the 9th IEEE Winter Conference on BCI. Track 3 focuses on imagined speech decoding (closed vocabulary) for intuitive BCI communication. The dataset was designed to evaluate whether EEG signals recorded while subjects imagine specific words can be classified into discrete categories.

It contains recordings from 15 subjects, each performing five distinct imagined speech classes: *"Hello"*, *"Help me"*, *"Stop"*, *"Thank you"*, and *"Yes"*. For every single subject, multiple sessions were recorded, with each class containing 70 trials, each trial comprising 795 EEG samples covering the time window from -500 ms to 2600 ms relative to the stimulus onset. EEG signals were acquired with a 64-channel cap following the 10–20 international system. Ground and reference channels were placed on the Fpz and FCz, at a sampling rate of 256 Hz. Data were provided in three subsets: training, validation, and test. The training set contains 300 trials (60 per class), the validation set 50 trials (10 per class), and the test set 50 trials (10 per class). In terms of array dimensions, this corresponds to $4500 \times 64 \times 795$ samples for training, $750 \times 64 \times 795$ for validation, and $750 \times 64 \times 795$ for testing. This information has been summarized in Table 3.1.

Table 3.1: Summary of BCI Competition 2020 Track 3 dataset specifications.

Feature	**Description**
Classes	5 imagined speech commands (*Hello, Help me, Stop, Thank you, Yes*)
Subjects	15 participants
Trials per class	70 trials per class, per subject
Samples per trial	795 (time window: -500 ms to 2600 ms)
Sampling rate	256 Hz
Channels	64 EEG electrodes (10–20 system)
Array dimensions	Train: $4500 \times 64 \times 795$, Val: $750 \times 64 \times 795$, Test: $750 \times 64 \times 795$

It is important to note that the BCI Competition 2020 Track 3 corpus is released in raw form, without any preprocessing applied to the EEG recordings. As illustrated in Figure 3.1, ULF drifts and power-line artifacts remain present in the signals, leaving preprocessing choices entirely up to the researcher.

File Format Note

From a technical standpoint, the training and validation files are stored in `MAT` v7.2 format, while the test set, uploaded later, uses `MAT` v7.3. This format mismatch introduces a practical issue: v7.3 files require `HDF5`-compatible readers (e.g., `h5py` in Python), while v7.2 files can be read directly with standard MATLAB or `scipy.io.loadmat`. Without accounting for this difference, attempts to load the test data may result in parsing errors.

3.2 Model architectures

In addition to the baseline architectures discussed in the state-of-the-art section, we also tested two modified versions of the Diff-E model. The rationale for including these variants was not to pursue architectural innovation or performance gains, but to examine whether the influence of preprocessing is consistent across models with different inductive biases. Attention mechanisms were incorporated as a representative alternative to the original convolutional blocks, given their increasing relevance in sequence modeling. By contrasting the original Diff-E design with these attention-augmented variants, we aim to determine whether architectural changes alter the models' susceptibility to artifacts, particularly ULF components, or whether the impact of preprocessing remains fundamentally independent of the architecture in imagined speech EEG decoding.

3.2.1 Proposed variants of Diff-E

3.2.1.1 Diff-E with a 1-D ViT encoder–decoder

In this variant of Diff-E, the convolutional autoencoder is replaced by a Transformer-based encoder–decoder. Transformers [33] are sequence modelling architectures that process a set of input vectors (tokens) through layers that combine two components: (i) multi-head self-attention, which allows each token to exchange information with all others, and (ii) a position-wise feed-forward network. A token can be understood as a fixed-length vector representation of a unit of information in the input sequence; in natural language processing this would typically be a word or subword, while in our EEG adaptation each token corresponds to a temporal patch covering P consecutive samples across all C EEG channels. This design avoids the sequential bottleneck of recurrent networks and allows modelling dependencies across arbitrarily long contexts, which is particularly valuable for EEG signals where relevant patterns may be separated in time.

The self-attention mechanism works by projecting the token sequence $\mathbf{Z} \in \mathbb{R}^{(N+1)\times d}$, where N is the number of patch tokens and d is the embedding dimension of each token, into three parallel spaces:

$$\mathbf{Q} = \mathbf{Z}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{Z}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{Z}\mathbf{W}_V, \tag{3.1}$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d_k}$ are learned parameters. Intuitively, $\mathbf{Q}$ (queries) represent the information each token is looking for, $\mathbf{K}$ (keys) represent what each token can offer, and $\mathbf{V}$ (values) are the actual content to be transferred. The attention score between token i and token j is computed as the scaled dot product between $\mathbf{q}_i$ and $\mathbf{k}_j$:

$$\alpha_{ij} = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_k}}, \tag{3.2}$$

and normalised across all j using the softmax function. These weights α_{ij} determine how much information token i aggregates from token j:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\alpha)\,\mathbf{V}. \tag{3.3}$$

This mechanism is applied in parallel across multiple *heads*, enabling the model to capture different types of relationships between tokens.

While the original Diff-E encoder extracts local temporal patterns through convolution, here we adopt the Vision Transformer (ViT) [34] adapted to 1-D EEG. In the standard ViT, an image is divided into fixed-size patches, each flattened and linearly projected into a d-dimensional token. In our 1-D adaptation, an EEG trial $\mathbf{X} \in \mathbb{R}^{C\times L}$ is split along the temporal axis into N non-overlapping segments (patches) of length P:

$$N = \frac{L}{P}. \tag{3.4}$$

Each patch $\mathbf{X}_{[k]} \in \mathbb{R}^{C\times P}$, where C is the number of EEG channels and P the number of time samples per patch, is flattened into a one-dimensional vector using the $\text{vec}(\cdot)$ operator, which stacks all $C \times P$ elements into a single column of size $CP \times 1$. This vector is then linearly projected into the d-dimensional embedding space of the Transformer:

$$\mathbf{e}_k = \mathbf{W}_{\text{emb}}\,\text{vec}(\mathbf{X}_{[k]}) + \mathbf{b}_{\text{emb}}, \quad \mathbf{e}_k \in \mathbb{R}^d, \tag{3.5}$$

where $\mathbf{W}_{\text{emb}} \in \mathbb{R}^{d\times CP}$ and $\mathbf{b}_{\text{emb}} \in \mathbb{R}^d$ are learnable parameters.

Two additional elements are standard in ViTs and are also used here. First, a learnable class token $\mathbf{e}_{\text{cls}} \in \mathbb{R}^d$ is prepended to the sequence before attention is applied. This token does not correspond to any specific patch; instead, it acts as a global summary vector that aggregates information from all other tokens through self-attention [35]. After the final encoder layer, the class token is fed to the classification head. Second, because attention is permutation-invariant, we add a learnable positional embedding $\mathbf{p}_k \in \mathbb{R}^d$ to each token to encode the temporal order of the patches and allow the model to distinguish their original positions in the sequence. In this expression, $\mathbf{e}_k \in \mathbb{R}^d$ denotes the embedding vector of the k-th patch token obtained from the patch embedding stage, $\mathbf{p}_k \in \mathbb{R}^d$ is a learnable parameter vector specific to the position k in the sequence, and $\tilde{\mathbf{e}}_k \in \mathbb{R}^d$ is the resulting position-aware token representation:

$$\tilde{\mathbf{e}}_k = \mathbf{e}_k + \mathbf{p}_k. \tag{3.6}$$

Here the addition is performed element-wise, injecting position-dependent information into the content representation of each token.

We choose a Transformer for this architecture because it introduces a different inductive bias from convolutional models: instead of relying solely on local receptive fields, it can model long-range temporal relationships between distant parts of the trial in a content-adaptive manner.

The ViT formulation is particularly suitable for this adaptation because the patch embedding stage naturally converts a multichannel EEG signal into a sequence of fixed-length tokens, making it directly compatible with the self-attention mechanism and allowing us to explore whether the influence of preprocessing differs in a globally attentive model compared to the original convolutional Diff-E.

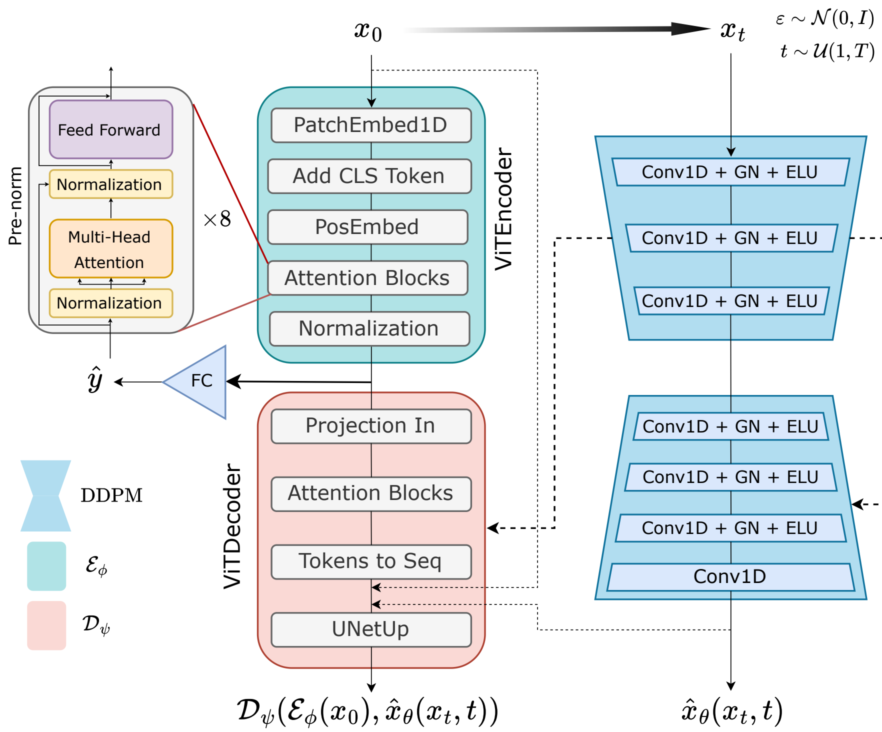

Figure 3.2: Architecture of the Transformer Diff-E.

To integrate the ViT into Diff-E without altering the overall generative–discriminative pipeline, we replaced the convolutional encoder–decoder with a Transformer-based encoder and decoder, while preserving the original skip connections[1] used in the CAE version. In the original Diff-E, intermediate feature maps from the encoder are passed directly to corresponding layers in the decoder to recover fine-grained temporal detail during reconstruction. In our adaptation, the patch tokens produced by the ViT encoder are decoded back into a sequence of the original length, and these reconstructed features are fused with the preserved skip connections from the diffusion model's intermediate stages, following the same alignment strategy as in the CAE. This design ensures that the ViT variant remains functionally interchangeable with the base Diff-E, allowing a fair comparison of preprocessing effects while controlling for changes in the rest of the architecture.

[1]A known risk of skip connections is that they may provide a direct copy of the input. Thus an average pooling before concatenation is applied.

3.2.1.2 Diff-E with axial attention and squeeze–excitation modulation

In this second modification, the original U-Net backbone of Diff-E is retained, but the residual convolutional blocks are augmented with two mechanisms: (i) axial self-attention and (ii) squeeze–excitation (SE) modulation. These additions aim to enrich the feature representation by modelling long-range dependencies along different axes and adaptively reweighting channels based on their relevance to the task.

3.2.1.2.1 Axial self-attention. To complement the convolutional processing in the U-Net blocks, we integrate an axial self-attention module [36], which factorises the attention operation[2] into two sequential passes along orthogonal axes of the EEG tensor $\mathbf{X} \in \mathbb{R}^{B\times C\times T}$:

- **Temporal self-attention:** Here, the sequence dimension is the temporal axis T, and each channel c is processed independently. This design allows each time step to attend to all others within the same electrode, capturing long-range temporal dependencies. In our implementation, we optionally restrict the receptive field to a temporal window $W < T$ for efficiency; when W is set to T, the attention is global. A feed-forward network and residual connections are applied after attention, following the Transformer block design [33].
- **Channel self-attention:** Here, the sequence dimension is the set of EEG channels C, and attention is computed independently for each time step. This allows the model to learn inter-channel dependencies, potentially reflecting spatial correlations or functional coupling between electrodes. The implementation applies a projection across channels, computes channel-wise attention, and rescales each channel's activation using learned weights. This re-weights channels according to their relevance at each stage.

3.2.1.2.2 Squeeze–excitation modulation. After axial attention, each residual block incorporates a SE module [37] to recalibrate the importance of each EEG channel. While axial attention captures dependencies across time and channels, its output treats all channels equally in the subsequent processing. The SE block introduces a complementary mechanism that explicitly models channel-wise relevance, enabling the network to emphasise electrodes carrying discriminative information and to down-weight those dominated by noise or irrelevant activity.

Formally, given a feature map $\mathbf{X} \in \mathbb{R}^{B\times C\times T}$, the SE block first squeezes the temporal dimension by global average pooling producing a descriptor $\mathbf{z} \in \mathbb{R}^{C}$ summarising the overall activation of each channel. This descriptor passes through a small bottleneck MLP with reduction ratio r, and the resulting weights $\mathbf{w} \in (0,1)^{C}$ modulate the original feature map in

[2] By factorising the attention into temporal and channel passes, we preserve the ability to capture global context along both dimensions while reducing the computational complexity from $\mathcal{O}((CT)^2)$ to $\mathcal{O}(C^2 + T^2)$.

the excitation step:

$$\tilde{\mathbf{X}}_{c,t} = w_c \cdot X_{c,t}. \tag{3.7}$$

By design, axial attention models relationships between channels and time steps, while the SE block operates as a content-dependent gating mechanism within the channel dimension. This combination lets the model capture global spatio–temporal context while reweighting EEG channels, which is useful in decoding since some electrodes carry more relevant information and noise levels can vary across trials and subjects.

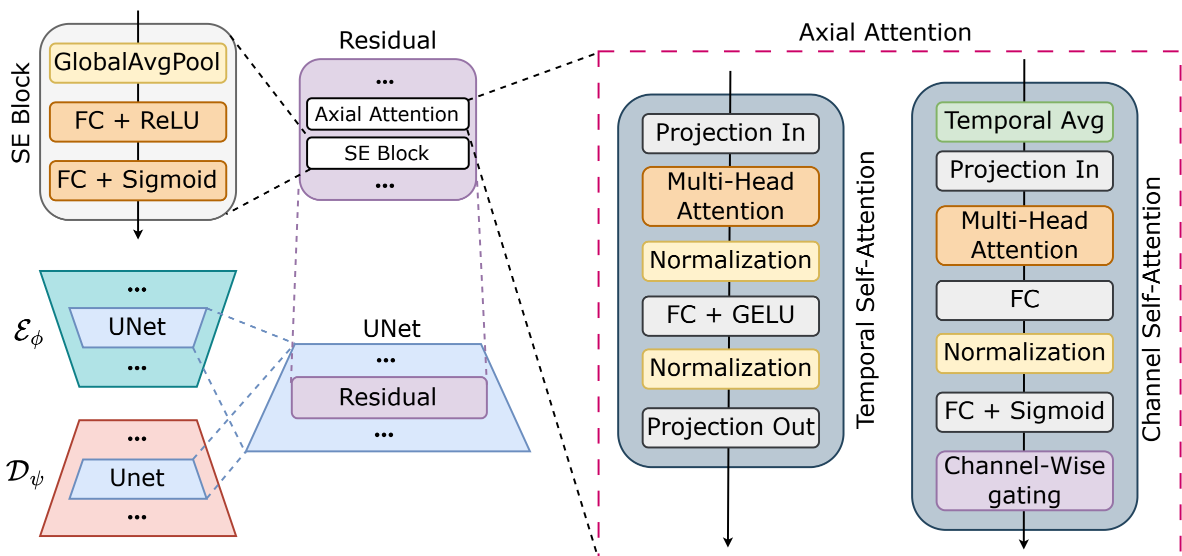

Figure 3.3: Architecture of the Diff-E with attention layers.

3.3 Experimental Protocols

To rigorously evaluate the impact of preprocessing and architectural choices, we designed a set of controlled experimental protocols. These define how data are partitioned, how preprocessing components are isolated through ablation, and how different model families are benchmarked under consistent conditions. The goal is to ensure that performance differences can be attributed to methodological factors rather than to uncontrolled variability in the evaluation setup.

We adopted two evaluation regimes: (i) within-subject, where training and testing were restricted to data from the same participant, and (ii) cross-subject, where models trained on a subset of participants were evaluated on held-out individuals. The former quantifies subject-specific decoding performance, while the latter assesses generalization across subjects.

In order to systematically examine the role of preprocessing, we designed an ablation study. Ablation studies are commonly used in machine learning to determine the contribution of individual components by selectively removing or adding them. In this work, the goal is to evaluate the risks of omitting standard preprocessing steps, as such omissions may allow spurious artifacts to remain in the data and bias decoding performance. Concretely, we compare model accuracy when individual steps are applied in isolation versus when they are removed from the full pipeline.

Mathematical definition: We can formally define our ablation study as follows. Let the raw EEG signal be denoted as $X \in \mathbb{R}^{C \times T}$ with C channels and T time samples. We define the set of preprocessing operators as

$$\mathcal{P} = \{\text{Band-pass}, \text{Notch}, \text{CAR}, \text{ICA}, \text{Baseline}\},$$

where each element $p \in \mathcal{P}$ represents a transformation $p : \mathbb{R}^{C \times T} \to \mathbb{R}^{C \times T}$. For each operator $p \in \mathcal{P}$, two complementary conditions were considered:

- X_p: the signal obtained by applying only the operator p to X, i.e. $X_p = p(X)$.
- $X_{\setminus p}$: the signal obtained by applying all operators in $\mathcal{P}$ except p, i.e. $X_{\setminus p} = (q_1 \circ \cdots \circ q_{|\mathcal{P}|-1})(X)$ with $q_i \in \mathcal{P} \setminus \{p\}$.

This design allows us to quantify the individual impact of each step, both when applied in isolation and when removed from the complete pipeline. In doing so, we can measure how strongly decoding performance depends on each preprocessing component and how much risk is introduced by not applying them.

In total, eight models were evaluated in the ablation study: MLP, a standard CNN, EEGNet, ShallowConvNet, DeepConvNet, the base Diff-E architecture, and the two modified versions of Diff-E incorporating attention blocks and transformer layers. To evaluate performance, we monitored a broad set of metrics capturing different aspects of classification quality. Alongside raw accuracy, we report balanced accuracy, macro-F1, macro-precision, macro-recall, and one-vs-one ROC-AUC. Agreement-based measures include Matthews correlation coefficient (MCC) and Cohen's kappa. For interpretability, confusion matrices were also computed. All results are compared against a random baseline, allowing us to quantify the true gain achieved by each model and preprocessing configuration beyond chance performance.

To investigate which aspects of the DC component drive class separability, we performed explainability analyses on the SVM model rather than on deep architectures. Given that the SVM with an RBF kernel operates directly on static DC features, its decision process can be interpreted through feature-level importance rather than convolutional filters or temporal maps.

First, we computed one-way ANOVA tests across classes for each channel, quantifying their relative contribution to discriminability in terms of F-scores and associated p-values. Second, we performed post-hoc pairwise t-tests with Bonferroni correction, complemented by effect size estimates (Cohen's d), to characterize which classes diverged most strongly within each channel. Third, we visualized per-class distributions using histograms and boxplots, providing an interpretable view of how DC offsets shift across conditions. Fourth, we applied SHAP (SHapley Additive exPlanations), a model-aware framework that attributes to each channel its average marginal impact on classifier outputs, thus capturing nonlinear and multivariate dependencies that ANOVA cannot detect. Finally, spatial topoplots were generated to illustrate both absolute DC levels and relative class-specific deviations (ΔDC), highlighting consistent spatial patterns across subjects.

Together, these analyses offer complementary perspectives: (i) statistical, via ANOVA, t-tests, and effect sizes; (ii) descriptive, through distribution plots; (iii) model-aware, through SHAP importance scores; and (iv) spatial, through topographical mapping. Importantly, the results consistently showed that only a subset of electrodes carries strong class-dependent DC shifts, and that these offsets—captured either as mean differences or as multivariate patterns—are the key factor exploited by the SVM for classification.

3.4 Computational Environment

All experiments were executed on *Magerit*, the high-performance computing (HPC) cluster hosted at CESVIMA (Centro de Supercomputación y Visualización de Madrid). Unlike a personal workstation, an HPC cluster is composed of hundreds of interconnected compute nodes designed to operate in parallel, making it possible to tackle large-scale scientific workloads that would be impractical on local hardware. Magerit is one of the largest academic computing facilities in Spain, supporting research across multiple disciplines.

Access to the cluster is performed remotely through secure shell (SSH), where users connect to dedicated login nodes. These nodes are not intended for computation but serve as gateways to prepare jobs, configure environments, and submit tasks to the scheduler. The cluster uses the SLURM workload manager, a queueing system that allocates jobs to compute nodes according to requested resources and availability. Jobs are typically submitted with the `sbatch` command, where the user specifies resource requirements (number of CPUs, GPUs, memory, and wall-clock time). SLURM then manages the queue and assigns jobs to nodes, enforcing fair usage and efficient resource distribution. Our experiments were run using PyTorch on compute nodes equipped with NVIDIA A100[3] GPUs. Each training job was allocated one GPU, four CPU threads, and 16 GB of host memory.

[3] The NVIDIA A100 is based on the Ampere architecture and offers up to 80 GB of HBM2e high-bandwidth memory, delivering over 2 TB/s of memory bandwidth. It provides peak FP16 tensor performance of up to 624 TFLOPS and supports Multi-Instance GPU.

Results and Discussion

4.1 Raw vs preprocessed

We conducted a series of experiments to evaluate the effect of preprocessing on the decoding performance of EEG signals. Specifically, we trained all models on data from three individual subjects (ID 2, ID 5, and ID 9) as well as on the pooled dataset containing all subjects. These subjects were not deliberately chosen for any specific property but rather selected in an approximately random manner to avoid introducing bias into the analysis and to prevent cherry-picked cases. For each configuration, training was repeated with three different random seeds (7, 21, and 42) to account for initialization variability. The seeds were likewise selected in a similarly arbitrary manner, following the same approach as with the subject choice, with the sole aim of providing representative variability. This setup allowed us to assess both subject-specific performance and generalization across subjects.

Preprocessing Protocol

Since the dataset is provided in an already epoch-based format, our preprocessing procedure was applied at the subject level. Specifically, we concatenated the training, validation, and test splits for each subject, applied the full preprocessing pipeline on the resulting continuous signal, and then re-split the data according to the original partitions. This choice is motivated by the assumption that recordings for a given subject were acquired within the same session. Applying preprocessing consistently across concatenated splits ensures that operations such as baseline correction, filtering, or normalization are not biased by partition boundaries. If preprocessing were applied independently to each split, discrepancies in reference levels or filter initialization could introduce artificial differences between train, validation, and test sets, compromising the validity of the evaluation.

In this section, we focus on the direct comparison between the raw signals and the fully preprocessed signals. The latter include the complete preprocessing pipeline described in Section 2.2, while the raw condition preserves the dataset in its original form. The figures present the classification accuracy obtained across the different models for both the raw and the fully preprocessed datasets. Each bar plot corresponds to a specific experimental setting: Subject 2, Subject 5, Subject 9, and the pooled dataset including all subjects. A horizontal dashed line at 20% (0.2) accuracy is included in all figures, indicating the random baseline of the five-class classification task.

Across all conditions, models trained on raw data consistently outperform those trained on preprocessed data (Figure 4.1). The performance drop introduced by preprocessing is particularly evident in the subject-dependent experiments (Appendix A), where the accuracy

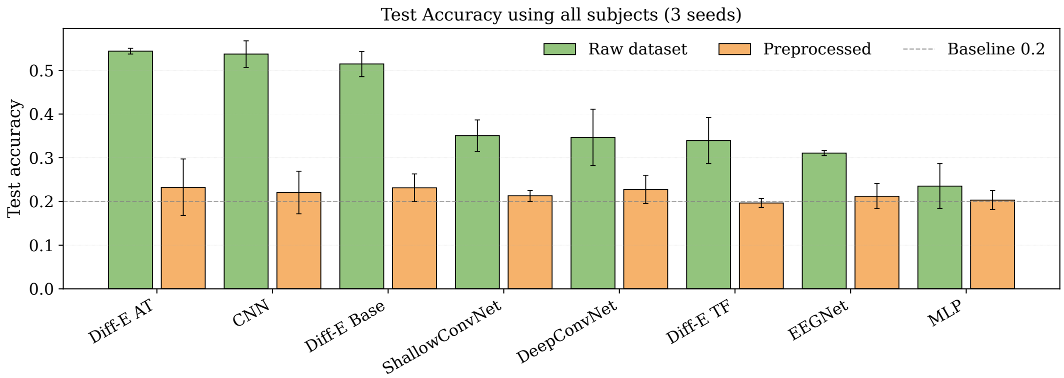

Figure 4.1: Classification accuracy across models for the raw and fully preprocessed datasets, aggregated across all subjects. Error bars represent variability across the three random seeds (7, 21, and 42).

gap between raw and preprocessed inputs is more pronounced. In contrast, for the pooled inter-subject case, the difference remains present but is noticeably smaller. Among the evaluated architectures, the standard CNN and the Diff-E variant with attention layers emerge as the best-performing models across most conditions. Overall, these results suggest that the preprocessing pipeline applied to this dataset reduces discriminative information relevant for subject-dependent decoding, while offering no clear benefit in the inter-subject setting. To evaluate whether preprocessing has a measurable effect on classification performance, we additionally performed paired t-tests between the raw and fully preprocessed conditions. These tests were applied separately for each subject and for the combined dataset.

Table 4.1 summarises the paired comparison between raw and preprocessed EEG signals across subjects and models. For each configuration, test accuracy values obtained under three independent seeds were paired, and the mean difference (Preprocessed–Raw) was evaluated using a paired t-test with FDR[1] correction. Confidence intervals (95%) and effect sizes were also computed.

Across all subjects and models, accuracies obtained from raw signals consistently surpassed those from the preprocessed pipeline. Mean differences ranged from -0.10 to -0.63, with confidence intervals excluding zero in almost all cases, indicating a clear degradation when preprocessing was applied. Paired t-tests confirmed these differences as statistically significant after FDR correction ($p < 0.05$). Effect sizes were very large (Cohen's $|d_z| > 2$ in most cases), highlighting the practical relevance of the observed performance gap. The consistency across multiple architectures, from shallow CNNs to transformer-based Diff-E variants, demonstrates that the negative impact of preprocessing is not model-specific but a general phenomenon

[1] All reported p-values were adjusted using the Benjamini–Hochberg False Discovery Rate (FDR) correction. This procedure controls the expected proportion of false positives across multiple hypothesis tests, providing a less conservative but more powerful alternative to the Bonferroni correction.

across the dataset. Only isolated cases (e.g., MLP on the pooled set) yielded inconclusive results, with confidence intervals overlapping zero.

Table 4.1: Comparison between raw and preprocessed data across subjects. Values are mean test accuracy (±95% CI) over three seeds. Effect size is reported as Cohen's d_z, and p_{FDR} denotes the paired t-test p-value after FDR correction. Negative d_z indicates degraded performance with preprocessing.

Exp.	Model	Raw	Preproc	d_z	p_{FDR}
sub2	CNN	0.733 ± 0.031	0.247 ± 0.013	-10.5	0.009
sub2	DiffE_AT	0.693 ± 0.047	0.253 ± 0.020	-7.3	0.013
sub5	CNN	0.833 ± 0.030	0.213 ± 0.012	-31.0	0.011
sub5	DiffE_AT	0.813 ± 0.045	0.247 ± 0.020	-7.0	0.013
sub9	CNN	0.660 ± 0.042	0.307 ± 0.024	-8.5	0.011
sub9	DiffE_AT	0.687 ± 0.049	0.347 ± 0.021	-4.3	0.023
all	CNN	0.537 ± 0.028	0.220 ± 0.011	-28.5	0.007
all	DiffE_AT	0.544 ± 0.036	0.232 ± 0.015	-10.9	0.010

4.2 Ablation Study

To assess the contribution of each individual preprocessing step, we performed an ablation study by systematically adding a single component P (X_p) to the raw data X. The resulting performance was then compared against the baseline obtained with the unprocessed raw dataset. This protocol isolates the marginal effect of each preprocessing stage on the final classification accuracy.

Figure 4.2 illustrates the change in test accuracy (Δ Test Acc.) across different models when different steps of preprocessing were individually introduced. Positive values indicate an improvement relative to the raw baseline, whereas negative values reveal a detrimental effect.

The results show that most preprocessing operations tend to reduce classification accuracy when applied in isolation to the raw dataset. In particular, the addition of baseline correction and band-pass filtering consistently led to the largest decreases in performance across multiple models. Other steps, such as notch filtering or ICA, had comparatively smaller but still generally negative impacts. These findings suggest that the models, when trained directly on raw signals, exploit low-frequency and noise-related components that are partially removed by conventional preprocessing stages.

In a complementary analysis, we evaluated the effect of removing each preprocessing step P ($X_{\setminus p}$) from the complete pipeline X_{full} . This approach quantifies the relative contribution of each component when operating within the full preprocessing chain. In this setting, a positive

Δ Test Accuracy indicates that omitting the step improves model performance, while negative values suggest that the step is beneficial.

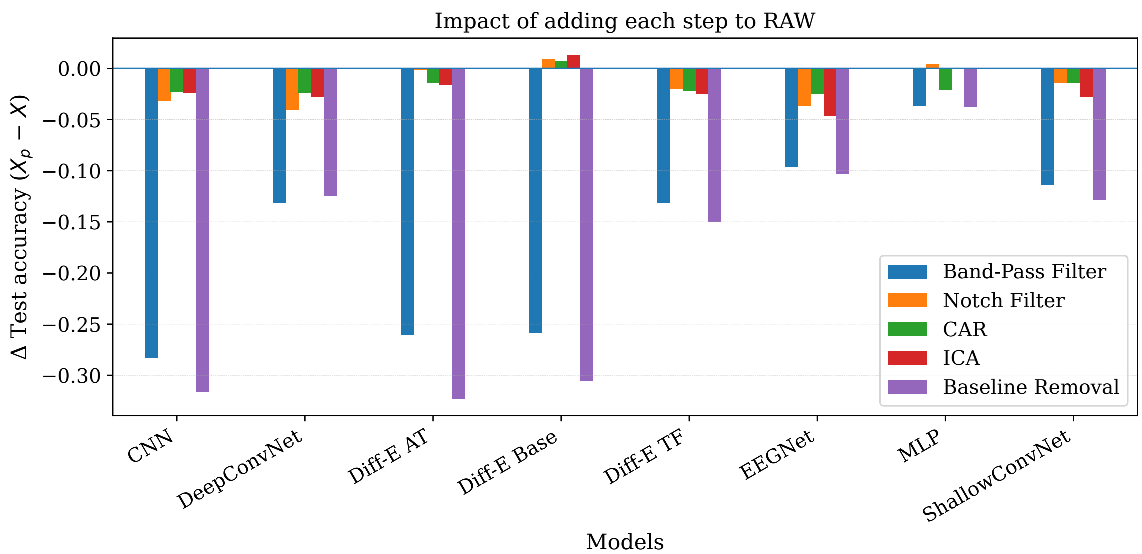

Figure 4.2: Impact of adding each preprocessing step individually to the raw data X_p on test accuracy across all evaluated models. Values correspond to Δ Test Accuracy with respect to raw X.

Figure 4.3 summarises the results. Compared to the X_p analysis, where accuracy drops reached up to -0.3 points, the effects observed in the $X_{\backslash p}$ setting are considerably smaller. The scale of variation is markedly reduced, indicating that once the full preprocessing pipeline is in place, removing individual steps barely alters the final accuracy.

Interestingly, the baseline correction show consistent patterns across the two protocols. In the X_p experiments, adding this step strongly reduced performance, while in the $X_{\backslash p}$ experiments its removal similarly led to relative gains. This symmetry suggests that the discriminative power of the models is linked to signal components that these steps suppress, ULF range.

It is worth noting that the baseline correction exhibits a non-linear, order-dependent behavior. When it is added on top of RAW, performance degrades, which we attribute to an unstable baseline estimate in the presence of strong ULF drift. In contrast, within the full pipeline, removing it can also reduce accuracy. This seemingly paradoxical pattern is explained by interaction effects: once ULF components are attenuated by the band-pass filter and channel-wise structure is regularized by CAR/ICA, baseline correction acts as a stable centering step that improves inter-trial consistency. Therefore, it is detrimental in isolation but beneficial in combination with the other preprocessing operations.

Taken together, X_p and $X_{\backslash p}$ experiments show that the largest performance changes are linked

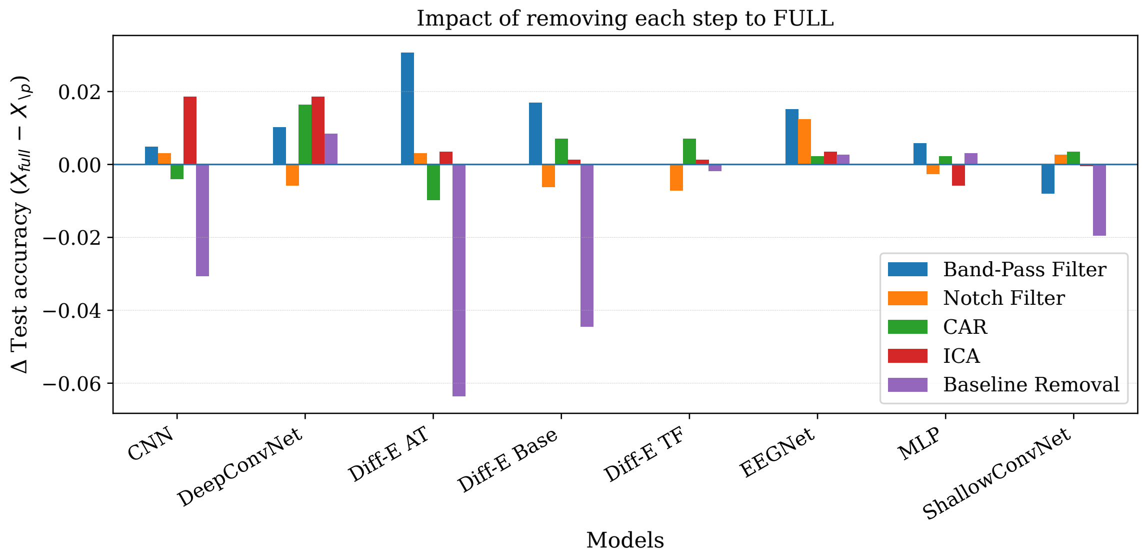

Figure 4.3: Impact of removing individual preprocessing steps from the full pipeline $X_{\setminus p}$ on test accuracy across all evaluated models. Values correspond to Δ Test Accuracy with respect to the full pipeline X_{full}.

to the band-pass filter and baseline correction. In both protocols, these steps systematically reduce accuracy, indicating that the models exploit information removed by them. This strongly suggests that ULF components, typically suppressed by standard preprocessing, contain discriminative features that drive classification performance. The role of ULF activity will be examined in detail in the next section.

4.3 Isolation of Ultra-Low Frequency Components

To investigate the origin of the ULF information seemingly exploited by the models, we conducted a frequency cutoff sweep. For each cutoff X, the signal was split into two complementary bands: the residual component below X Hz and the complementary band above X Hz and separate models were trained on each. This approach allows us to assess which spectral regions contribute most strongly to classification performance.

The choice of the minimum cutoff frequency is constrained by the acquisition parameters of the dataset. With a sampling rate of $f_s = 256$ Hz and trial length of $N = 795$ samples (corresponding to $T \approx 3.1$ s), the spectral resolution of a discrete Fourier transform is $\Delta f = f_s/N \approx 0.32$ Hz. This value represents the smallest resolvable frequency step, and thus the lowest meaningful cutoff frequency that can be implemented. Frequencies below this threshold cannot be reliably isolated given the trial duration. The Nyquist frequency of the dataset is

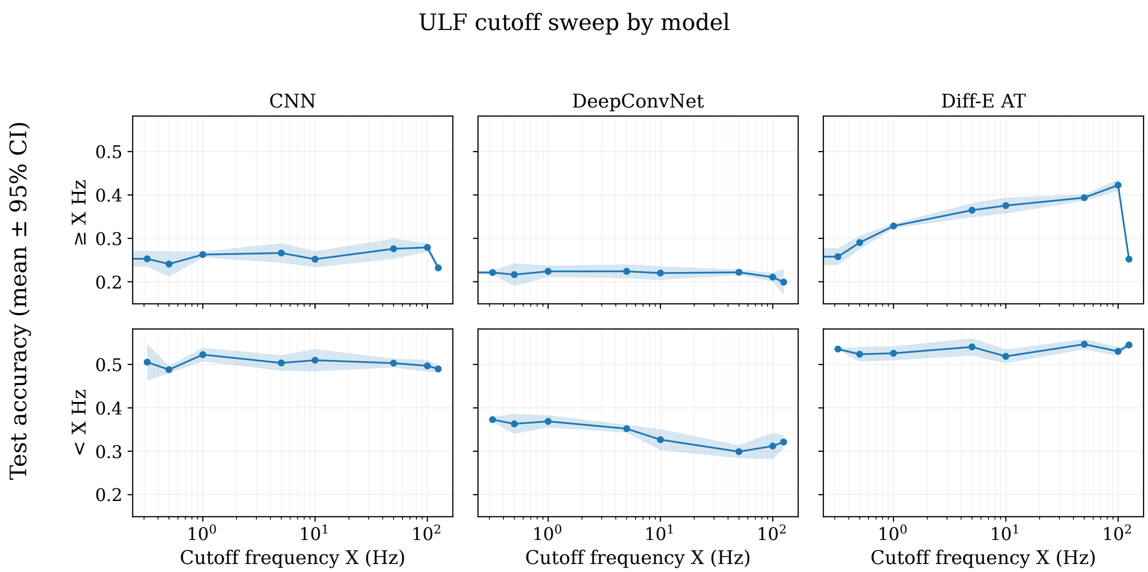

Figure 4.4: ULF cutoff sweep for CNN, DeepConvNet, and Diff-E AT. Residual accuracy peaks at 0.32 Hz ($\sim$0.5 vs. $\sim$0.2 complementary), while Diff-E AT improves with higher cutoffs, consistent with high-γ activity.

$f_{\text{Nyquist}} = f_s/2 = 128$ Hz, which defines the upper spectral bound.

Taking these constraints into account, we selected cutoff frequencies of 0.32, 0.5, 1, 5, 10, 50, 100, and 125 Hz. The lowest cutoff (0.32 Hz) aligns with the spectral resolution limit and provides the closest possible approximation to isolating the DC component. Figure 4.4 illustrates the results for three representative models, while the complete set of models is reported in the Appendix.

Figure 4.4 reports the cutoff sweep for three representative models. For CNN, increasing the cutoff frequency does not yield any clear improvement when training on the complementary band ($\geq X$ Hz). Similarly, the residual component ($< X$ Hz) maintains a stable accuracy across the sweep, suggesting that the discriminative information exploited by this model is not strongly distributed across distinct frequency ranges. Nevertheless, a remarkable phenomenon appears at the lowest cutoff: when splitting at 0.32 Hz, the complementary band yields an accuracy of only $\sim$0.2, whereas the residual band achieves $\sim$0.5. This sharp contrast between DC (0 Hz) and 0.32 Hz indicates that critical information exploited by the models is concentrated precisely in the ultra-low frequency range.

Another interesting trend emerges in the Diff-E AT model (and similarly in its baseline variant). Unlike CNN and DeepConvNet, the complementary-band accuracy gradually increases as the cutoff frequency rises. We hypothesize that this behavior stems from the greater representational capacity of the Diff-E architecture, which allows it to exploit patterns at higher frequencies. As discussed in Section 3, high-γ activity (70–150 Hz) is known to carry overt speech information, and prior work with Diff-E explicitly leveraged this band. Our interpretation is that, as the cutoff approaches this region, Diff-E begins to capture genuine neuronal correlates of speech rather than spurious low-frequency drifts, leading to the observed improvement.

These findings remarks the importance of the frequency range between 0 and 0.32 Hz. The sharp performance difference observed when isolating this interval strongly suggests that the critical source of spurious discriminative information may reside in the direct current (DC) component itself. In the following section, we investigate this hypothesis in detail, assessing whether the DC offset is indeed the origin of the problem.

4.3.1 DC Component as a Potential Source of Bias

Having identified the critical contribution of the [0–0.32] Hz interval, we now turn our attention to the DC component. By definition, the DC offset of an EEG signal corresponds to the zero-frequency component, i.e., the constant baseline around which the signal oscillates. From a neurophysiological perspective, the DC level is not expected to carry meaningful neural information, as it should remain constant across trials. Neural information arises from oscillatory activity, and by definition, DC does not oscillate. However, in this dataset, the DC level varies systematically with the class.

Instead, it is generally attributed to slow drifts, electrode polarization, or other hardware-related artifacts. While such artifacts can be present, they are not expected to exhibit systematic differences across experimental conditions. Therefore, the presence of class-discriminative patterns in the DC component would be highly unusual and, given the lack of supporting evidence in the literature, more plausibly linked to non-neuronal influences, although a neural contribution cannot be entirely ruled out.

To probe this issue, we explicitly isolate the DC component and use it as the sole input for classification. If models achieve comparable accuracy to those trained on broadband EEG, this would indicate that systematic low-frequency components, rather than broadband neural activity, play a dominant role in driving performance. In this section, we evaluate this hypothesis by training a simple linear classifier (SVM) exclusively on DC values and comparing its results to those of deep learning models. To further substantiate this concern, we conducted a targeted experiment focusing on a single participant (Subject 1), in order to more clearly illustrate the effect. Nonetheless, we emphasize that the same phenomenon was observed consistently across all participants. For this analysis, we extracted only the DC component of the EEG signals by computing the average across time for each channel, yielding a dataset of size 300×64 for training.

A SVM with an RBF kernel was trained using exclusively these DC values as input features.

Importantly, we repeated the experiment with multiple random seeds and obtained identical results in all cases, ruling out variability due to random initialization. The best validation accuracy achieved was 64%, and the corresponding test accuracy reached 70%. The resulting confusion matrix is shown in Figure 4.5.

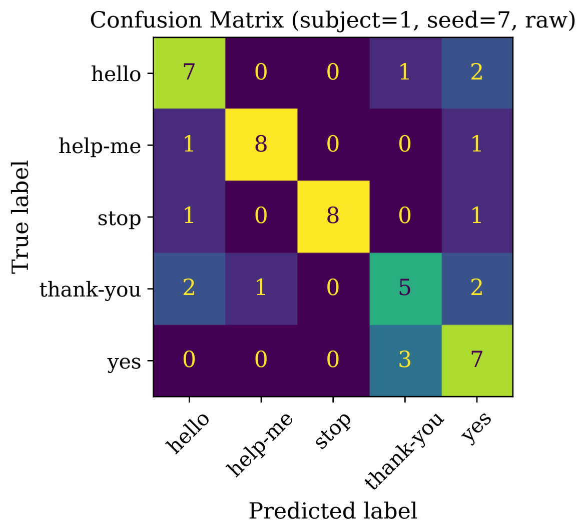

Figure 4.5: Confusion matrix obtained for Subject 1 using only the DC component as input to an SVM with RBF kernel, reaching an accuracy of 70% in test.

These results provide compelling evidence that the discriminative performance observed in previous models may be largely driven by spurious information contained in the DC component, rather than by genuine neural activity. In fact, a simple classical classifier, when trained solely on DC values, was able to match the performance of state-of-the-art deep learning approaches, thereby highlighting the risk of bias introduced by this non-neuronal source of variance.

To investigate the origin of this discriminative DC component, we performed a one-way ANOVA across classes for each channel. The use of ANOVA is justified because our goal was not to model the full non-linear separability between classes, but rather to test whether simple mean differences in DC level exist across class labels at the level of individual channels, although the data as a whole are not linearly separable. The results, presented in Figure 4.6, identify a subset of electrodes with particularly high F-scores, suggesting that the distribution of DC values varies systematically across conditions in these locations. This provides evidence that only a limited number of sensors contribute disproportionately to the spurious class separability.

To further characterize these channels, we plotted the class-wise distributions of DC values, as shown in Figure 4.7. Several electrodes exhibit clear class-dependent shifts in their mean DC levels, while others show overlapping but still distinguishable patterns. For each electrode, a one-way ANOVA was applied across the five classes, and the channels with the highest

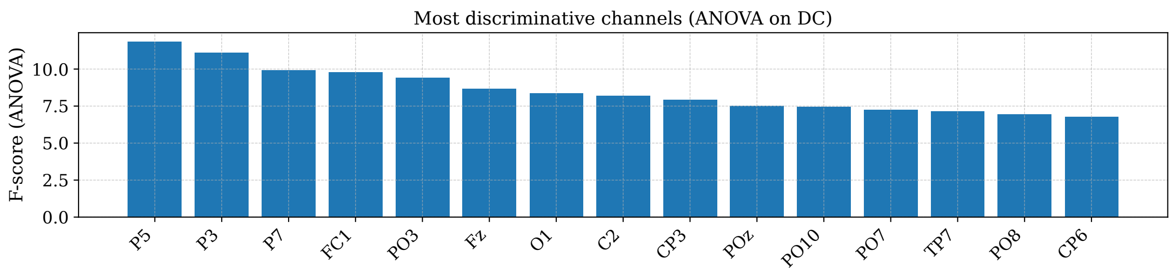

Figure 4.6: ANOVA F-scores across channels for the DC component, highlighting the electrodes with the highest discriminative power.

F-scores were selected for further post-hoc testing. The five most discriminative electrodes were located primarily over the left parietal and occipital regions (P5, P3, P7, PO3) together with one fronto-central site (FC1). All of these channels yielded highly significant main effects ($p \ll 0.001$), confirming that the class-wise DC means differ systematically.

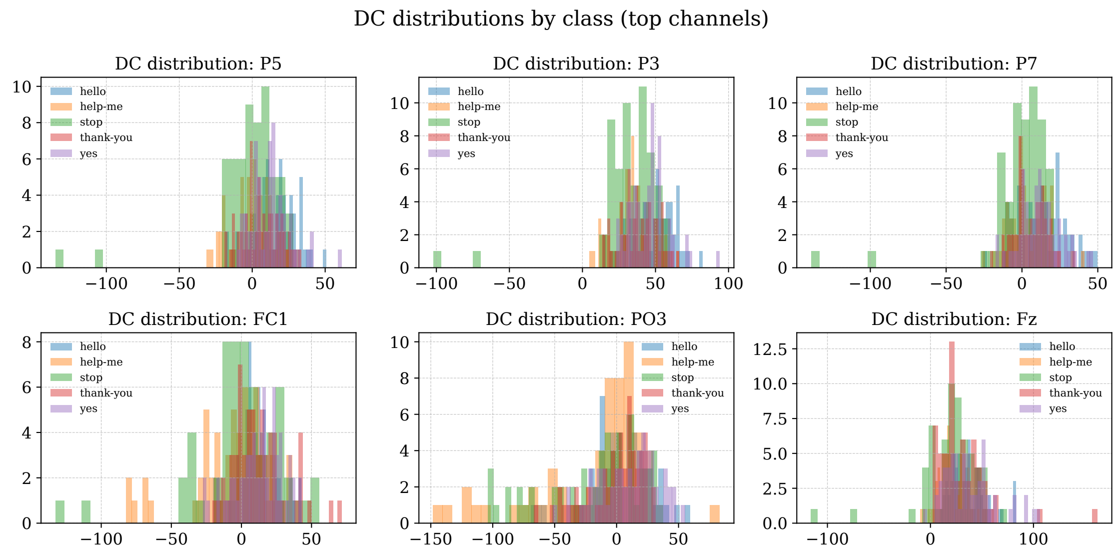

Figure 4.7: Class-wise histograms of DC values for the channels with the highest ANOVA scores. Distinct distributions across conditions highlight that DC offsets, rather than oscillatory features, account for the observed discriminability.

Pairwise t-tests with Bonferroni correction revealed consistent patterns across electrodes: the words *hello* and *yes* tended to exhibit higher DC offsets, while *help-me* and *stop* showed lower values. The class *thank-you* generally occupied an intermediate range, except at FC1 where it clustered with the higher group. Effect sizes were moderate to large (Cohen's $d \approx 0.6$–1.0), indicating that these differences are not only statistically significant but also of substantial magnitude.

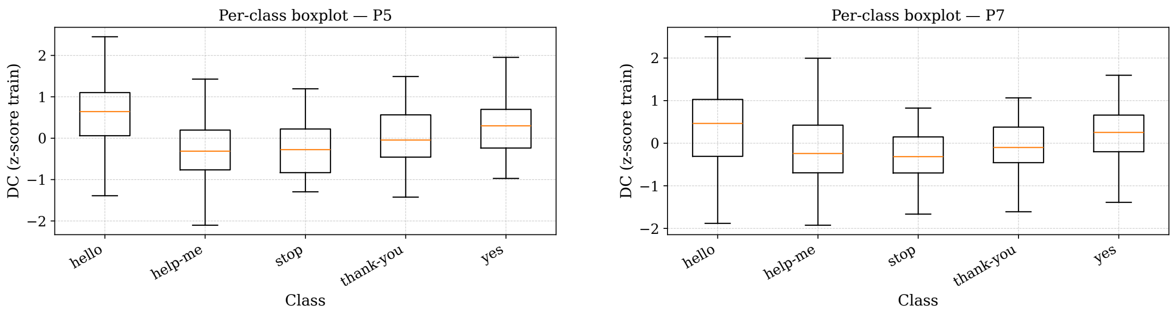

Figure 4.8: Per-class distributions of DC values (z-scored) for two of the most discriminative channels (P5 and P7). The observed systematic offsets across classes illustrate how static DC differences can drive apparent separability.

As a complementary analysis, we generated topographical maps for each class, shown in Figure 4.9. The top row depicts the absolute DC distribution obtained by averaging across trials for each condition, thereby illustrating the raw class-specific offsets. In contrast, the bottom row shows the relative DC modulation for each class (ΔDC), computed by subtracting the global mean across all conditions. This representation highlights class-specific deviations while controlling for global drifts common to all classes. Consistent with the boxplots in Figure 4.8, distinct spatial patterns emerge: for example, *hello* and *yes* exhibit systematically higher offsets across parietal-occipital regions, whereas *help-me* and *stop* show lower relative values. These maps reinforce the evidence that the observed discriminability arises from structured DC shifts rather than dynamic neural oscillations.

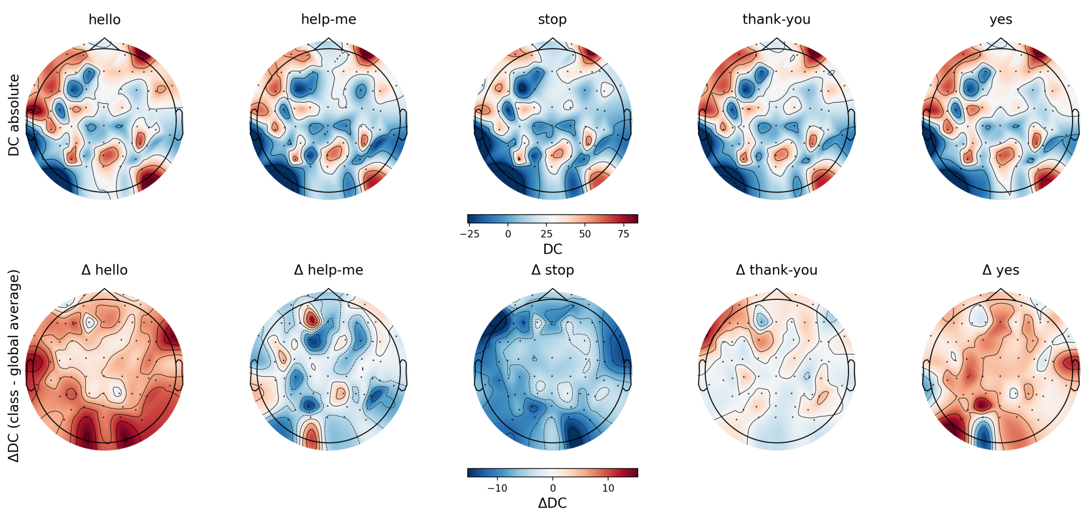

Figure 4.9: Topographical distribution of DC offsets across classes. Top row is the absolute DC mean per class. Bottom row is the class-specific ΔDC values relative to the global average.

While the ANOVA analysis provided evidence that a limited set of electrodes exhibit systematic class-dependent differences in their mean DC level, this approach is inherently univariate and linear, focusing on simple mean shifts per channel. To complement this perspective, we employed SHAP, which estimates the contribution of each feature to model predictions in a multivariate and model-aware fashion. SHAP is grounded in cooperative game theory and assigns each channel a Shapley value reflecting its average marginal impact on the classifier output across all possible feature coalitions. In this way, SHAP captures interactions and nonlinear effects that ANOVA cannot detect, thereby providing a fuller characterization of how the SVM RBF model exploits DC information across electrodes.

Figure 4.10 shows the SHAP summary plots for two representative subjects (Subject 5 and Subject 1) trained on DC-only data. Each horizontal bar corresponds to a channel, with colors indicating the contribution of different classes, and the x-axis representing the mean absolute SHAP value (i.e., the average magnitude of channel impact on predictions).

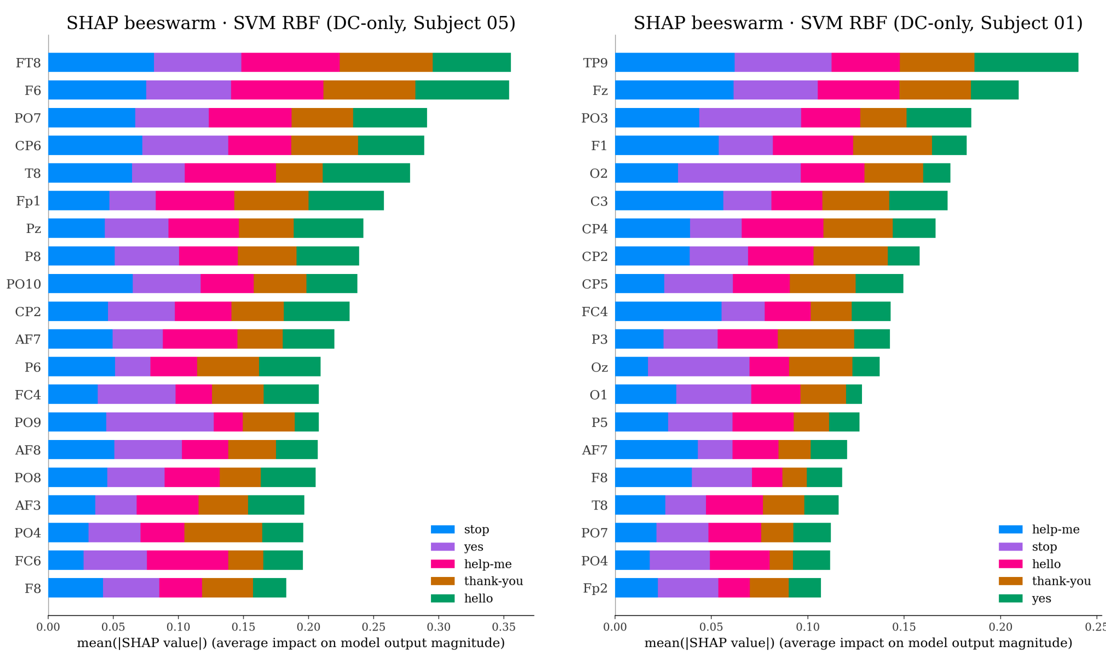

Figure 4.10: SHAP beeswarm plots for Subject 05 (left) and Subject 01 (right) obtained from an SVM RBF trained on DC-only data. Bars represent the average importance of each channel, with class contributions color-coded.

The comparison between ANOVA and SHAP highlights a key distinction. ANOVA identified a small subset of electrodes where the mean DC value differed significantly across classes, pointing to localized univariate effects. By contrast, SHAP revealed that the SVM RBF model does not rely exclusively on these channels but instead exploits broader, subject-specific patterns distributed across multiple electrodes. This indicates that the discriminative structure cannot be reduced to simple mean shifts in a few locations, but rather emerges from multivariate spatial combinations of DC offsets. Despite this variability, classification remains accurate, indicating that the discriminative structure is not tied to a fixed neurophysiological source but instead arises from spatially heterogeneous DC offsets that machine learning models can systematically exploit.

An alternative hypothesis for the origin of these DC shifts is that they may have arisen from the recording protocol itself. While the dataset documentation states that the word prompts were presented in randomized order, we considered the possibility that recordings might have been conducted in blocks, such that slow impedance drifts in certain electrodes could inadvertently encode class information. To test this, we sorted the trials by their mean DC value and inspected the class distribution across the sorted sequence, as shown in Figure 4.11. If block-wise acquisition were the source of the effect, we would expect a systematic clustering of classes along this ordering. However, no such ordering emerged, suggesting that the acquisition protocol is unlikely to explain the observed phenomenon.

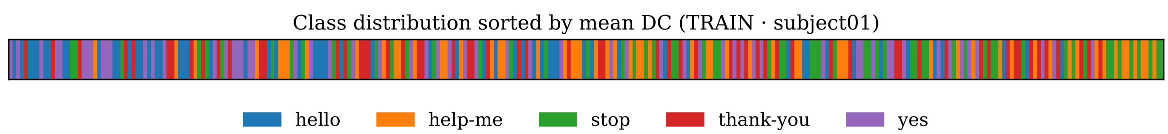

Figure 4.11: Class distribution for Subject 1 after sorting trials by their mean DC value. No clear block-wise structure can be observed, indicating that the acquisition protocol itself is unlikely to account for the systematic DC differences.

Moreover, we verified this behavior in two independent datasets. First, in the *Thinking Out Loud* [38] corpus, where no comparable class-dependent DC drift was present. Second, in an internally collected dataset from the Grupo de Bioingeniería y Telemedicina (GBT) at Universidad Politécnica de Madrid, which also involved silent speech recordings. In both cases, no evidence of systematic DC-related class separability was observed, reinforcing the interpretation that this effect is specific to the BCI Competition 2020 Track 3 dataset.

4.4 Discussion

Our analyses show that the DC component carries a substantial amount of class-discriminative information in the BCI Competition 2020 Track 3 dataset. When this component is retained, even simple classifiers can achieve high decoding accuracies, whereas removing it leads to a marked drop in performance. This pattern suggests that models trained on raw signals may rely heavily on slow offsets rather than on broadband neural activity. Notably, this effect disappears when applying standard preprocessing operations such as band-pass filtering or baseline correction, indicating that these steps suppress the discriminative contribution of the DC activity.

The phenomenon appears consistently across subjects and across all architectures tested. Although every model benefits from the presence of the DC component, those architectures that capture its presence most effectively, such as convolutional neural networks and diffusion-based approaches (Diff-E), show the largest performance decline once it is removed. This indicates that their advantage on raw data may stem in part from their ability to exploit spatially distributed slow drifts. At the same time, our results confirm that the effect is not confined to a trivial linear offset in a single electrode: the RBF SVM analysis suggests that classification depends on structured combinations of channel-wise shifts, which makes the effect difficult to detect by simple inspection.

The precise nature of these slow components remains unresolved. On the one hand, the literature does not report robust physiological signatures of silent or imagined speech in the DC range. On the other, slow cortical potentials are known to exist in EEG, and electrode or amplifier drifts may also contribute to systematic low-frequency structure. Our reordering analysis provided no evidence for block-wise recording artifacts, consistent with the experimental documentation, but this does not fully exclude other recording-related factors. The absence of similar class-dependent DC patterns in an independent dataset (*Thinking Out Loud*) further suggests that the effect may be specific to BCI2020 Track 3, although additional replications are needed before reaching firm conclusions.

Several, not mutually exclusive, hypotheses may explain the presence of class-correlated structure in the DC range. Physiological factors could play a role. Slow cortical potentials associated with sustained attention, task set, or speech imagery strategies might generate shifts over several seconds, while subtle modulations in subvocal muscle tone, respiration, or ocular drift specific to the imagined word could likewise contribute quasi-DC activity. Although such class-specific DC effects have not been robustly documented in the silent speech literature, the current evidence is not sufficient to entirely exclude a neural contribution. Recording-related mechanisms offer an alternative explanation. Electrode–skin polarization, impedance fluctuations, amplifier drift, or grounding interactions with the stimulus hardware may introduce slow offsets that become partially class-locked if certain conditions, such as uneven class distribution across time, sessions, or micro-blocks, are present.

Taken together, these findings highlight an unusual property of this dataset: classification performance is strongly dependent on the DC component, a signal structure whose physiological versus non-physiological origin remains uncertain. Rather than providing evidence of noise or of genuine neural decoding, our results point to the need for caution when interpreting high accuracies obtained on raw signals, and to the importance of clearly reporting and testing preprocessing choices in silent speech decoding research.

Conclusions and future work

5.1 Conclusions

The primary objective of this work was to examine the methodological implications of omitting preprocessing in EEG-based silent speech decoding. Our analyses confirmed that the BCI Competition 2020 Track 3 dataset contains strong class-discriminative structure in the ultra-low-frequency range, and in particular in the DC component. Consistent with **H1**, we found that this component alone allows both simple and complex models to reach accuracies comparable to those reported in the literature, highlighting that a substantial part of performance in this dataset is driven by slow offsets of unresolved origin. This finding directly addresses our first objective, showing that raw pipelines do not necessarily exploit broadband neural activity but may instead rely on low-frequency structures.

The second hypothesis **H2** predicted that standard preprocessing pipelines would mitigate the influence of these components. Our results support this claim: both band-pass filtering and baseline correction effectively removed class-dependent DC structure, resulting in decreased but potentially more physiologically grounded decoding performance. This observation meets our secondary objective of systematically comparing preprocessing configurations, as it demonstrates the decisive role of these steps in controlling for DC-related effects.

Hypothesis **H3** proposed that the impact of preprocessing would depend on the dataset and the model architecture. Our experiments confirmed this: while all models lost accuracy after DC removal, the degree of decline varied. Architectures such as CNNs and Diff-E, which appear particularly effective at capturing DC-related structure, showed the sharpest performance drops. This suggests that architectural differences modulate the extent to which models exploit slow components. Finally, **H4** suggested that the DC component could be systematically characterized as a central factor in model decisions. Our statistical analyses (ANOVA across channels) and classification experiments with SVMs support this view, showing that classification is not explained by trivial linear offsets in a single electrode, but by structured combinations across multiple channels.

From a broader perspective, this work positions itself as a methodological contribution. By contrasting raw and preprocessed pipelines across eight architectures, we have clarified the extent to which reported accuracies in BCI2020 Track 3 depend on ultra-low-frequency structure. Importantly, although our additional analyses on the *Thinking Out Loud* dataset did not reveal similar class-dependent DC effects, this single replication is insufficient to exclude a possible physiological contribution. As such, the interpretation of DC offsets as purely non-neural must remain tentative.

5.2 Future Work

Several directions emerge from our findings. First, replication on additional corpora is essential to determine whether class-discriminative DC patterns are specific to BCI2020 Track 3 or represent a broader phenomenon in silent speech EEG. Second, targeted control analyses should be conducted to test alternative hypotheses about the origin of these effects, including per-trial baselining, alternative referencing schemes, and the use of auxiliary signals such as EMG or EOG to track potential peripheral contributions. Third, future datasets should adopt acquisition protocols that minimize the likelihood of DC-related confounds, for example by carefully monitoring electrode impedances, amplifier stability, and potential cross-talk with stimulus hardware. Fourth, explainability analyses should be extended to quantify the relative contribution of genuine oscillatory features versus slow drifts across spatial regions and frequency bands. Finally, architectures such as Diff-E that retain some robustness after DC removal warrant further development under stricter artifact control, as they may represent promising candidates for identifying physiologically plausible correlates of silent speech.

Ethical, economic, societal and environmental aspects

6.1 Introduction

This thesis explores the influence of EEG preprocessing strategies on the performance of machine learning models for silent speech decoding, using the open-access dataset from the BCI Competition IV (Track 3, 2020). The work is allocated at the intersection of neurotechnology and artificial intelligence systems. Its motivation comes from the need to design decoding pipelines that are not only technically accurate but also ethically responsible, socially beneficial, economically accessible, and environmentally sustainable.

By improving the understanding of how preprocessing affects decoding performance, this research contributes to the development of BCI systems that could be deployed in real-world scenarios. Such systems hold significant potential to help individuals with severe speech or motor impairments, directly aligning with the United Nations Sustainable Development Goals (SDGs) [39], particularly SDG 3 (*Good Health and Well-Being*), SDG 9 (*Industry, Innovation and Infrastructure*), and SDG 10 (*Reduced Inequalities*).

6.2 Description of Relevant Impacts Related to the Project

From an ethical perspective, the work is grounded in the use of publicly available, anonymized EEG data obtained under informed consent. No new data collection was carried out, and all processing complied with established research ethics guidelines [40]. This ensures that participant privacy is preserved and that the findings can be reproduced transparently, consistent with FAIR data principles [41].

Societally, the potential impact is profound. Silent speech BCIs could enable direct neural communication for people who have lost the ability to speak, restoring a level of autonomy that current assistive devices cannot always provide. At the same time, the ability to decode internal speech introduces the emerging challenge of cognitive privacy [42], making it essential to consider governance frameworks before large-scale deployment.

Economically, the results of this thesis could help optimize computational pipelines, allowing a high-performance decoding without excessive preprocessing overhead. This is really important for the design of low-cost BCI systems deployable outside research laboratories, which could reduce barriers to access in low-resource contexts.

In **environmental** terms, the project has a minimal ecological footprint, as it reuses an existing dataset rather than requiring new data acquisition. Computational experiments were conducted on shared high-performance computing infrastructure, avoiding unnecessary retraining and selecting model architectures that balance accuracy with computational efficiency. This approach is consistent with calls for sustainable AI practices [43].

6.3 Detailed Analysis of One of the Principal Impacts

One of the most significant considerations arising from this research is the ethical and legal challenge of safeguarding cognitive privacy. The signals processed in silent speech decoding are not mere biometric identifiers; they can contain information about a person's mental states or intentions. Even when anonymized, EEG data carries an inherent sensitivity because advances in analysis methods could extract unintended features in the future.

If deployed in real-world settings, these systems must therefore adhere to strict principles: informed consent that fully communicates potential risks; data minimization, collecting only what is necessary for the intended task; and secure storage and transmission of neural data, using strong encryption to prevent unauthorized access. Furthermore, usage should be restricted to clearly defined, beneficial purposes, excluding applications in surveillance, coercion, or commercial profiling. Without such safeguards, public trust in neurotechnology could be severely undermined, and promising innovations might face social resistance.

This aligns closely with SDG 3, by protecting the dignity and well-being of users; SDG 9, by ensuring that neurotechnological innovation is responsible and trustworthy; and SDG 10, by preventing misuse that could exacerbate inequalities.

6.4 Conclusions

From an ethical and societal perspective, this thesis underscores the necessity of embedding privacy protection and responsible governance into the design of neurotechnological tools from the outset. From an economic perspective, it demonstrates that efficient preprocessing strategies can reduce computational costs and expand accessibility, thus increasing the likelihood of adoption in diverse contexts. From an environmental perspective, it exemplifies sustainable research practice by reusing open data, leveraging shared infrastructure, and prioritizing computational efficiency.

Incorporating sustainability criteria and aligning the work with the SDGs has provided an additional layer of value, ensuring that the technical findings are not isolated from their real-world context. By situating this research within a broader ethical, societal, and environmental framework, it reinforces the principle that engineering innovation should not only advance performance metrics but also contribute to the creation of technology that is inclusive, responsible, and sustainable.

Financial Budget

Table 7.1 presents the estimated project costs, including personnel (320 h of AI Engineer work at 40 €/h), depreciation of a personal computer, and the fair market value of HPC GPU compute (NVIDIA A100) provided in-kind by the university. General costs (15%), industrial benefit (6%), and consumables for printing and binding are also included. All values are in euros with 21% VAT where applicable.

PERSONNEL COSTS (direct costs)		Hours	Hourly cost	Total
AI Engineer (ML/Neurotech)		320	40 €	12.800,00 €
MATERIAL RESOURCES COSTS (direct costs, DC)	**Purchase price**	**Months of use**	**Depreciation (years)**	**Total**
Personal computer (incl. software)	1.500,00 €	3	5	**75,00 €**
HPC GPU compute (A100), fair value	1.543,50 €	3	—	**1.543,50 €**
	TOTAL COSTS OF MATERIAL RESOURCES			**1.618,50 €**
GENERAL COSTS (indirect costs, IC)	15%	on DC		**2.162,78 €**
INDUSTRIAL BENEFIT	6%	on DC+IC		**994,88 €**
CONSUMABLES				
Printouts				**20,00 €**
Binding				**100,00 €**
SUBTOTAL BUDGET				**17.696,16 €**
VAT			21%	**3.716,19 €**
TOTAL BUDGET				**21.412,35 €**

Table 7.1: Budget table filled with project-specific figures. GPU compute included as fair-value market cost.

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Annexes

A Raw vs. Preprocessed per Subject

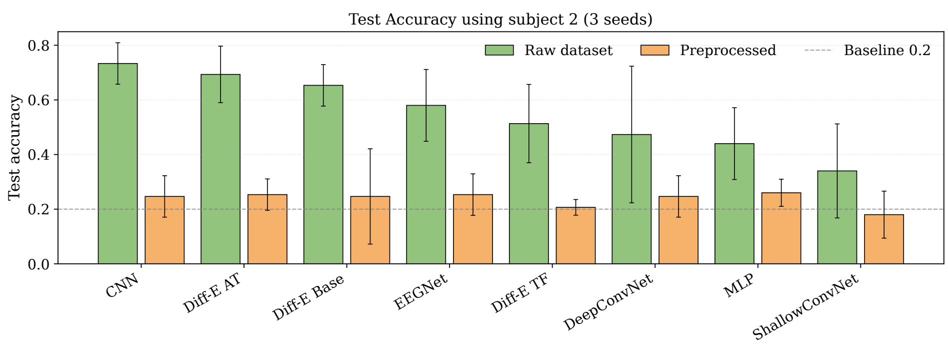

Figure 7.1: Classification accuracy across models for the raw and fully preprocessed datasets, shown for subject 2. Error bars represent variability across the three random seeds (7, 21, and 42).

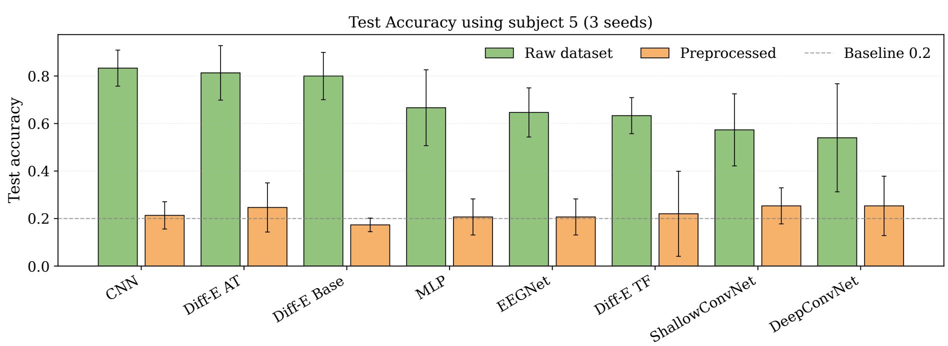

Figure 7.2: Classification accuracy across models for the raw and fully preprocessed datasets, shown for subject 5. Error bars represent variability across the three random seeds (7, 21, and 42).

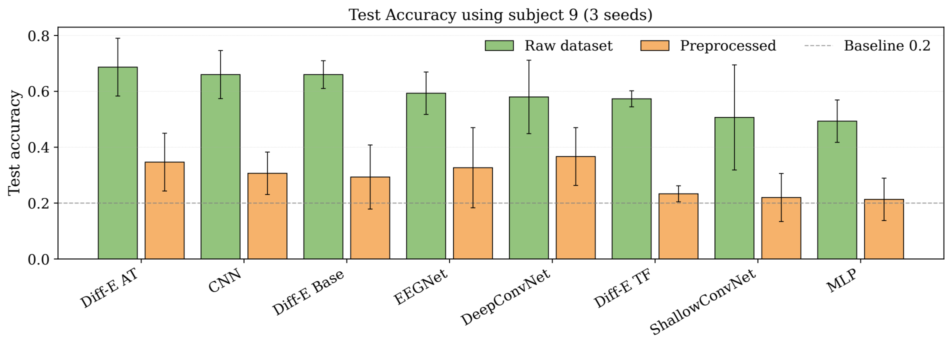

Figure 7.3: Classification accuracy across models for the raw and fully preprocessed datasets, shown for subject 9. Error bars represent variability across the three random seeds (7, 21, and 42).

B Raw vs. Preprocessed (split tables)

Table 7.2: Raw (P0) vs. Preprocessed (P1). Test accuracy for Subject 2. Values are mean accuracy across three seeds with uncertainty expressed as ± half-width of the 95% CI of the paired difference (P1–P0). Effect size is Cohen's d_z (paired), and p_{FDR} is the paired t-test p-value after FDR correction. Negative d_z indicates that P1 underperforms P0.

Model	**Raw (P0)**	**Preproc (P1)**	$\boldsymbol{d_z}$	$\boldsymbol{p}_{\mathrm{FDR}}$
DiffE_mod	0.653 ± 0.160	0.247 ± 0.160	-6.3	0.0146
DiffE_TF	0.513 ± 0.160	0.207 ± 0.160	-4.8	0.0191
DiffE_AT	0.693 ± 0.149	0.253 ± 0.149	-7.3	0.0131
MLP	0.440 ± 0.179	0.260 ± 0.179	-2.5	0.0529
CNN	0.733 ± 0.115	0.247 ± 0.115	-10.5	0.0087
EEGNet	0.580 ± 0.076	0.253 ± 0.076	-10.7	0.0093
ShallowConvNet	0.340 ± 0.149	0.180 ± 0.149	-2.7	0.0483
DeepConvNet	0.473 ± 0.291	0.247 ± 0.291	-1.9	0.0787

Table 7.3: Raw (P0) vs. Preprocessed (P1). Test accuracy for Subject 5. Same conventions as in Table 7.2.

Model	**Raw (P0)**	**Preproc (P1)**	d_z	p_{FDR}
DiffE_mod	0.800 ± 0.125	0.173 ± 0.125	-12.5	0.0098
DiffE_TF	0.633 ± 0.201	0.220 ± 0.201	-5.1	0.0174
DiffE_AT	0.813 ± 0.201	0.247 ± 0.201	-7.0	0.0134
MLP	0.667 ± 0.099	0.207 ± 0.099	-11.5	0.0100
CNN	0.833 ± 0.050	0.213 ± 0.050	-31.0	0.0111
EEGNet	0.647 ± 0.050	0.207 ± 0.050	-22.0	0.0073
ShallowConvNet	0.573 ± 0.131	0.253 ± 0.131	-6.0	0.0151
DeepConvNet	0.540 ± 0.235	0.253 ± 0.235	-3.0	0.0393

Table 7.4: Raw (P0) vs. Preprocessed (P1). Test accuracy for Subject 9. Same conventions as in Table 7.2.

Model	**Raw (P0)**	**Preproc (P1)**	d_z	p_{FDR}
DiffE_mod	0.660 ± 0.160	0.293 ± 0.160	-5.7	0.0154
DiffE_TF	0.573 ± 0.050	0.233 ± 0.050	-17.0	0.0074
DiffE_AT	0.687 ± 0.199	0.347 ± 0.199	-4.3	0.0230
MLP	0.493 ± 0.131	0.213 ± 0.131	-5.3	0.0170
CNN	0.660 ± 0.104	0.307 ± 0.104	-8.5	0.0113
EEGNet	0.593 ± 0.207	0.327 ± 0.207	-3.2	0.0367
ShallowConvNet	0.507 ± 0.174	0.220 ± 0.174	-4.1	0.0239
DeepConvNet	0.580 ± 0.057	0.367 ± 0.057	-9.2	0.0104

Table 7.5: Raw (P0) vs. Preprocessed (P1). Aggregated test accuracy across all subjects. Same conventions as in Table 7.2.

Model	**Raw (P0)**	**Preproc (P1)**	d_z	p_{FDR}
DiffE_mod	0.515 ± 0.037	0.231 ± 0.037	-19.1	0.0073
DiffE_TF	0.340 ± 0.053	0.196 ± 0.053	-6.7	0.0137
DiffE_AT	0.544 ± 0.071	0.232 ± 0.071	-10.9	0.0100
MLP	0.235 ± 0.032	0.203 ± 0.032	-2.3	0.0610
CNN	0.537 ± 0.028	0.220 ± 0.028	-28.5	0.0065
EEGNet	0.311 ± 0.032	0.212 ± 0.032	-7.7	0.0128
ShallowConvNet	0.351 ± 0.025	0.213 ± 0.025	-13.8	0.0094
DeepConvNet	0.347 ± 0.049	0.228 ± 0.049	-6.0	0.0145

C ULF cutoff sweep

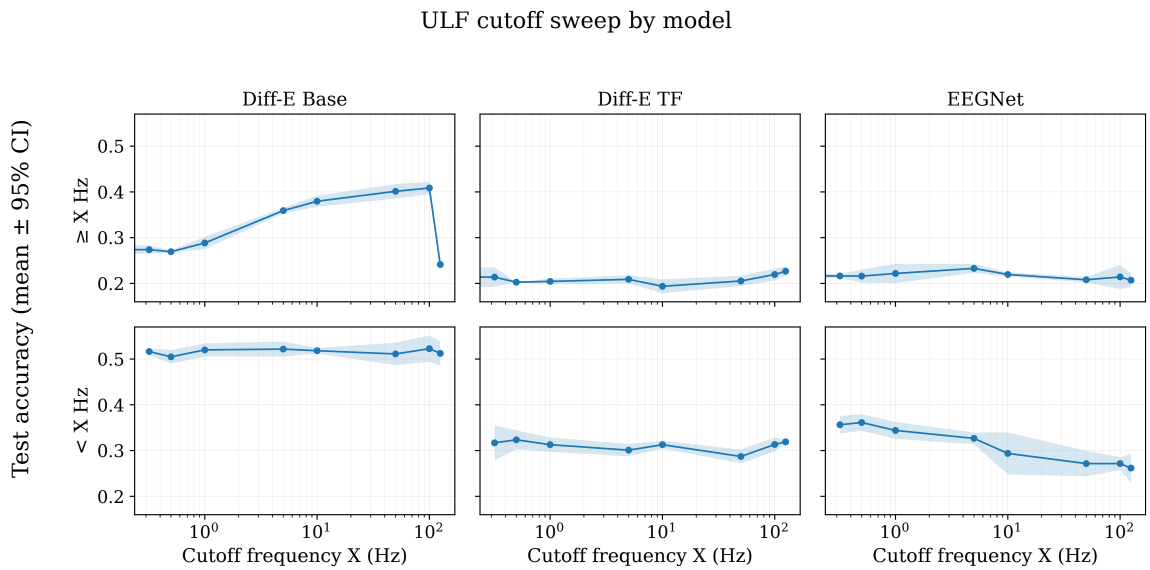

Figure 7.4: ULF cutoff sweep for Diff-E Base, Diff-E TF and EEGNet.

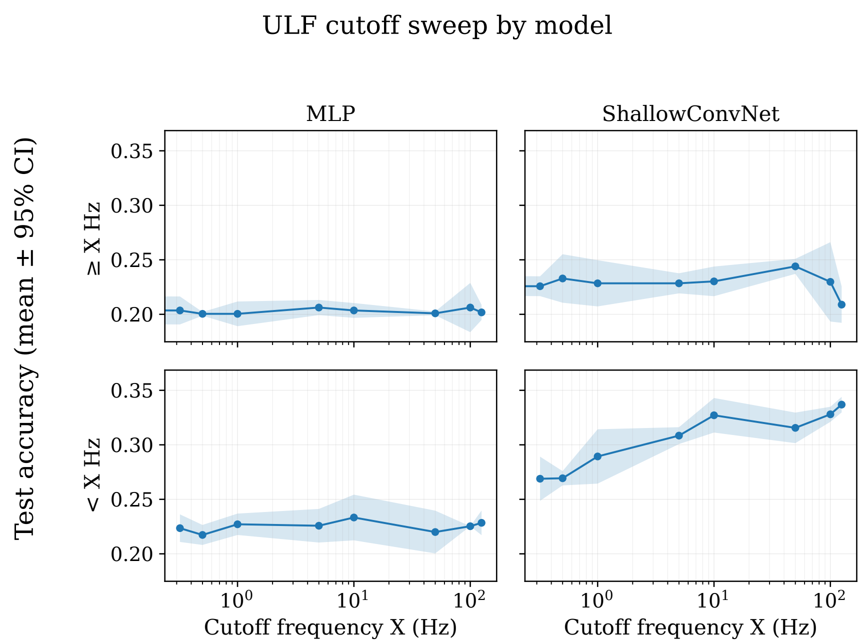

Figure 7.5: ULF cutoff sweep for MLP and ShallowConvNet.